Energy Conversion and Management: X 23 (2024) 100671

Contents lists available at ScienceDirect

Energy Conversion and Management: X

journal homepage: www.sciencedirect.com/journal/energy-conversion-and-management-x

Towards highly efficient solar photovoltaic thermal cooling by waste heat utilization: A review

Mena Maurice Farag [a,b], Abdul-Kadir Hamid [a,b], Maryam Nooman AlMallahi [c], Mahmoud Elgendi [c,d,e,*]

[a] Department of Electrical Engineering, College of Engineering, University of Sharjah, PO Box 27272, Sharjah, United Arab Emirates
[b] Smart Automation and Communication Technologies Research Center (SACT), Research Institute of Sciences and Engineering, University of Sharjah, PO Box 27272, Sharjah, United Arab Emirates
[c] Department of Mechanical and Aerospace Engineering, College of Engineering, United Arab Emirates University, Al Ain, United Arab Emirates
[d] National Water and Energy Center, United Arab Emirates University, Al Ain, P.O. Box 15551, United Arab Emirates
[e] Department of Mechanical Power Engineering and Energy, Faculty of Engineering, Minia University, Minia, Egypt

ARTICLE INFO

Keywords:
Adaptive and Robust Controlled Schemes
Predictive Cooling
PV/T
Sensors
Solar Energy
Optimization

ABSTRACT

Photovoltaic (PV) systems are popular for their reliability and zero fuel costs. However, only around 20 % of solar energy is converted into electricity, while the remainder is dissipated as waste heat. Excessive waste heat affects the lifespan of PV systems, leading to abnormal operating temperatures. In this notion, Photovoltaic-thermal (PV/T) systems are introduced to extract waste heat through various cooling techniques to harness electrical and thermal energies, demonstrating their capabilities through experimental and modeling techniques. Researchers have sought to develop optimized modeling techniques based on empirical, semi-empirical, and AI-based modeling for efficient execution of PV/T systems. This study reviews the current optimization developments in the PV/T systems, focusing on multiple numerical and experimental designs. Various cooling methods, including air, water, and phase change materials (PCM) with nanofluids, are examined for their promising contributions to electrical and thermal efficiency enhancement. Additionally, optimization methods have been investigated by incorporating automated processes into PV/T systems employing self-automation techniques. These processes aim to reduce the overall cost and establish a self-sustaining performance. Finally, the challenges and recommendations for future research for PV/T enhancement are highlighted.

1. Introduction

Greenhouse gas (GHG) emissions, global warming, and fossil fuel depletion require critical research attention to ensure a sustainable, eco-friendly, and clean future [1,2]. Renewable energy sources (RES), such as solar [3,4], wind [5], tidal [6], biomass [7], and geothermal energy [8], are progressively utilized to mitigate fossil fuel emissions [9]. Among all renewable energy resources, solar energy has seen exponential growth over the past two decades [10,11]. This is due to its zero fuel cost, high reliability, versatility, ease of installation, and low maintenance costs [12,13]. The solar market is built upon crystalline silicon technology, which is widely accepted for solar photovoltaic (PV) applications due to its relatively stable efficiency under varying temperature conditions, unlike traditional polycrystalline silicon-based photovoltaic cells.

Although PV technology has numerous advantages, it also faces significant challenges, including a shorter operational lifespan and higher electricity generation costs than thermal energy generation [14]. Additionally, a major critical issue in PV system generation is their low electrical efficiency since a substantial portion of the incident solar irradiance is dissipated as waste heat, leading to increased cell temperatures and decrement in electrical efficiency. This potential drawback necessitates advanced technologies that can effectively extract waste heat energy from PV systems to enhance their performance significantly [15,16].

Photovoltaic-thermal (PV/T) systems present a promising solution utilizing a broad solar energy spectrum, capturing light and heat [17]. A typical PV/T system comprises several aspects: a PV module, DC motors,

* Corresponding author at: Department of Mechanical and Aerospace Engineering, College of Engineering, United Arab Emirates University, Al Ain, United Arab Emirates.
E-mail address: mahgendi@uaeu.ac.ae (M. Elgendi).

https://doi.org/10.1016/j.ecmx.2024.100671
Received 22 February 2024; Received in revised form 27 June 2024; Accepted 21 July 2024
Available online 22 July 2024
2590-1745/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

collectors, coolant channels, and various coolants such as air, water, nanofluids, phase change materials (PCM), heat pumps, etc. [18–25]. Additionally, heat pumps are widely used in solar systems to enhance the system's efficiency [26–28]. The fluid in the heat pump system is heated using the solar heat that the solar collectors capture [29–32].

The performance of PV/T systems is primarily assessed through their electrical, thermal, and exergy efficiencies [33,34]. Electrical efficiency is highly sensitive to solar cell temperature since materials used in solar cells are highly subjected to temperature variations [35,36]. However, the power generation from PV/T systems can be unpredictable due to fluctuations in several parameters such as solar irradiance, geographical location, tilt angle, module temperature, and environmental conditions like wind direction, humidity, pressure, and dust accumulation [37]. The parameters vary over time, preventing PV modules from consistently operating at optimal conditions. Therefore, effective optimization techniques are crucial for enhancing the overall performance of PV/T systems [38,39].

Optimization of PV/T systems involve using advanced methods to maximize their efficiency, durability, and reliability. This includes enhancing electrical and thermal efficiency, reducing operational costs, and improving system performance. Multiple optimization techniques have been developed, from traditional modeling and simulation to advanced computational methods, artificial intelligence (AI), and intelligent technologies, as presented in Fig. 1.

The optimization techniques can be described as follows:

- Traditional modeling and simulation: These techniques involve mathematical modeling and numerical simulations, such as empirical and semi-empirical modeling and computational fluid dynamics (CFD), to predict the behavior of PV/T systems under different conditions. These models help understand the systems' electrical and thermal dynamics and identify optimal operational parameters for the cooling process [40,41].
- Artificial intelligence (AI) and machine learning (ML): AI and ML techniques have been widely adopted to analyze large datasets to identify patterns and predict system performance [42]. These methods can further optimize PV/T systems by predicting the optimal operating conditions and adjusting experimental parameters in real time to maintain peak efficiency.
- Smart methods and Internet of Things (IoT): IoT with PV/T systems enables real-time monitoring and control through sensors and wireless communication [43,44]. Smart methods involve adaptive frameworks that adjust cooling mechanisms based on real-time conditions, ensuring optimal performance under varying weather conditions [45,46].

Despite the significant potential of optimization techniques in PV/T systems, a comprehensive review of optimization approaches is uncommon. Therefore, this review aims to analyze these optimization approaches, highlighting their advancements and identifying areas for future research. This review systematically categorizes and analyzes these techniques, addressing the gaps in existing literature and offering valuable insights for research and practitioners.

The main contributions of this review are as follows:

- A thorough review of various optimization techniques applied to PV/T systems, including traditional and numerical modeling, AI/ML, and IoT-based techniques.
- The study discusses the state-of-the-art technologies in PV/T system optimization, providing insights into the latest innovations and advancements.
- The study identifies existing research gaps and proposes future directions to enhance the effectiveness of optimization techniques for PV/T systems.
- Practical recommendations are discussed for implementing advanced optimization methods in real world applications to improve the PV/T system performance.

This paper is organized as follows: Section 2 explores photovoltaic-thermal (PV/T) systems, assessing the influence of temperature on PV system performance and presenting various cooling techniques for waste heat extraction. Section 3 discusses potential technologies for adaptive cooling frameworks, reviewing the latest state-of-the-art. Section 4 addresses the challenges and limitations of current approaches, proposing future research directions. Finally, Section 5 concludes the paper, summarizing key findings and recommendations.

2. Photovoltaic-thermal systems

2.1. Impact of temperature on PV System's performance

Photovoltaic technology mainly depends on environmental parameters such as solar insolation and the module's operating temperature in electrical conversion [47]. However, under harsh climatic conditions, the evident impact of abnormal temperature is a significant factor since PV module technologies convert merely 20 % of solar insolation, and the remainder is dissipated as waste heat, leading to its consequent degradation [48,49].

PV modules are exposed to electromagnetic wave spectrums such as Infrared (IR) and Ultraviolet (UV), which constantly contribute to the PV module's lifespan degradation due to heat dissipation [50,51].

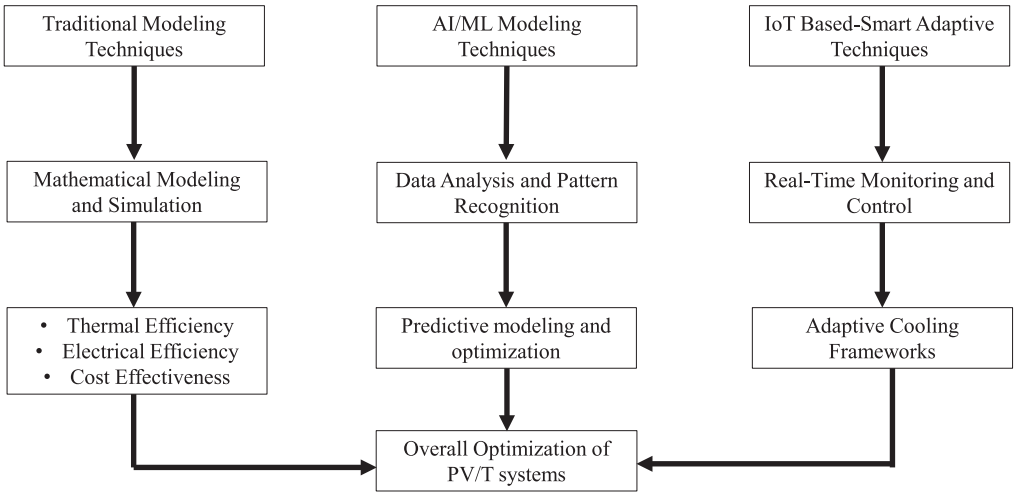


Fig. 1. Optimization approaches developed for PV/T systems illustrating different performance metrics for enhancement.

However, they are associated with a temperature coefficient (β) for power generation, symbolizing a degradation in %/°C with every 1 °C above the standard testing conditions (STC), i.e., 25 °C [52]. Various PV cell technologies vary regarding power degradation coefficient with respect to temperature [53].

Theoretical relations have been established within the scientific literature in equations (1)-(3) that define the dependence of crucial electrical parameters such as maximum power point (P_{mpp}), maximum operating voltage (V_{mpp}), and electrical conversion efficiency (η) on operating temperature, respectively [54]. The power generated at maximum power point (P_{mpp}) at a single distinct interval is interpreted through Eq. (1) as reported in [55]:

$$P_{mpp} = P_{STC} \times \frac{G}{G_{STC}} \times [1 + \beta_{STC}(T_{PV} - T_{STC})] \quad (1)$$

where the power at standard testing conditions is defined as P_{STC}, the solar insolation at actual and standard testing conditions is defined as G and G_{STC}, respectively, where G_{STC} is defined as 1000 W/m^2. The temperature coefficient for power defined for the PV module technology is defined as β_{STC}, where the coefficient represents a degradation in maximum power point as the temperature exceeds standard testing conditions, i.e., 25 °C, finally the operating temperature at actual and standard testing conditions is defined as T_{PV} and T_{STC}, respectively where T_{STC} is defined as 25 °C.

Similarly, the maximum operating voltage (V_{mpp}) at a single distinct interval is interpreted through Eq. (2) as reported in [56]:

$$V_{mpp} = V_{STC} \times \frac{G}{G_{STC}} \times [1 + \gamma_{STC}(T_{PV} - T_{STC})] \quad (2)$$

where the voltage at standard testing conditions is defined as V_{STC}, which coincides with the open circuit voltage defined for the PV module. γ_{STC} represents the temperature coefficient for voltage defined for the PV module technology, representing a decrement in voltage by %/1 °C as the operating PV temperature surpasses the prescribed STC conditions, i.e., 25 °C.

Finally, the electrical conversion efficiency is interpreted through Eq. (3) as reported in [57]:

$$\eta = \eta_{STC} \times [1 + \beta_{STC}(T_{PV} - T_{STC})] \quad (3)$$

Where η_{STC} is the electrical efficiency prescribed by the manufacturer at STC conditions. The impact of temperature degradation on crucial electrical parameters that define the overall performance of PV technologies can be illustrated in Fig. 2.

2.2. Utilization of photovoltaic-thermal systems

Photovoltaic-thermal (PV/T) systems have been introduced to enhance the overall performance of a PV system based on multiple active or passive cooling methods [58,59]. Cooling techniques allow waste heat extraction, which can be used for preheating applications [60,61]. Such a technique enhances the maximum power point and electrical conversion efficiency since the operating temperature is reduced [62]. These systems are considered Hybrid PV systems that extract both thermal and electrical energy, enhancing the overall performance of the PV system [63].

In principle, PV/T systems harness wasted heat produced by the PV cell to enhance the overall performance of any given PV system. Multiple techniques have been adopted to extract waste heat from PV cells [64–72]. Several types of cooling methodologies are utilized for PV/T systems such as passive cooling [73,74], active cooling [75,76], and hybrid cooling systems [77,78],. Therefore, PV/T systems generate both electrical and thermal energies, providing power stability and economic development of PV systems and their enhancement in large-scale applications such as grid-connected PV/T systems [79–81].

In this essence, PV/T systems are categorized as hybrid PV systems because of their capability to extract both electrical and thermal energies and boost the overall performance of any given PV energy system [82]. Hybrid PV/T systems harvest electrical (η_{el}) and thermal efficiency (η_{th}). The achieved efficiencies, when combined, form the overall performance of the PV/T system and are demonstrated as follows in Eq. (4) as reported in [83]:

$$\eta_{PVT} = \eta_{th} + \eta_{el} \quad (4)$$

The electrical efficiency (η_{el}) is typically derived via experimentation through Eq. (5) as reported in [84]:

$$\eta_{el} = \frac{P_{max}}{A_{PV} \times G} \quad (5)$$

Where P_{max} is the measured maximum power point at a single interval,

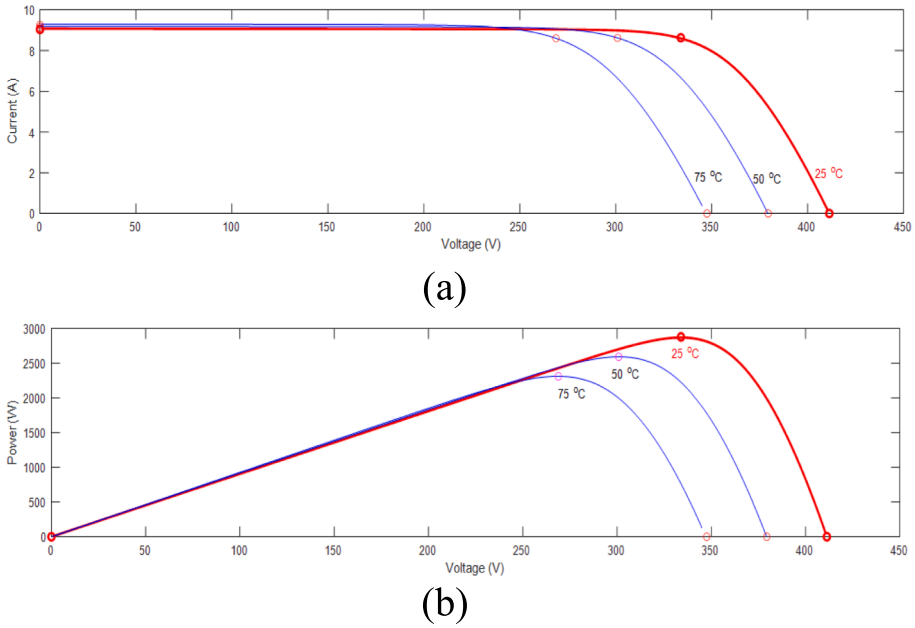


Fig. 2. Variation of Electrical Characteristics with module temperature simulated by MATLAB (a) I-V Characteristics (b) P-V Characteristics.

A_{PV} is the surface area of the PV module or array in m^2, and G is the measured solar irradiance in W/m^2.

In addition, the thermal efficiency (η_{th}) can be computed to determine the useful thermal energy extracted based on the cooling process [85]. It is calculated as shown in Eq. (6):

$$\eta_{th} = \frac{Q_u}{A_{PV} \times G} \tag{6}$$

Where Q_u represents the heat absorbed by the base fluid and is computed as follows through Eq. (7) as reported in [86]:

$$Q_u = \dot{m} C_p (T_o - T_i) \tag{7}$$

Where $\dot{m}$ is defined as the fluid mass flow rate in (kg/s), C_p is the specific heat capacity of the fluid (J/kg.K), and T_o and T_i are the outlet and inlet fluid temperatures (°C), respectively.

The reduction in thermal stress-based cooling methods enhances the longevity and lifespan of PV modules while enhancing the overall efficiency of the PV/T system. Several studies have reported the thermal heat pattern based on the PCM material added [87], as presented in Fig. 3 (i), air-based cooling [88], Fig. 3 (ii) water fluid [89] as presented in Fig. 3 (iii), showing the potential development of cooling methods. Additionally, the variation in heat extraction differs based on the location, ambient conditions, and the effectiveness of the employed cooling method. Moreover, heat pipes are extensively tested in various applications, including PV/T, vacuum tubes, and solar collectors [90].

2.3. Modeling and computation techniques for photovoltaic-thermal systems optimization

PV/T systems are emerging technologies combining electrical and thermal energy extraction benefits. Therefore, the modeling and computation of PV/T systems are essential to optimize their performance and understand their complex uncertainties and design conditions [91,92]. Mathematical models are developed to simulate performance parameters such as solar irradiance, ambient temperature, fluid flow, and electrical output. The derived mathematical models consider simultaneous heat and electricity generation, allowing the assessment of the overall efficiency and performance of PV/T under various operation conditions [93]. Regarding computational models, numerical methods and algorithms are implemented to enable system behavior predictions and analysis. Moreover, integrating modeling and computation methods facilitates the exploration of various design optimizations and the evaluating of different system configurations [94–97].

2.3.1. Computational fluid dynamics-based modelling technique

Computational fluid dynamics (CFD) models simulate the fluid flow, heat transfer capability, and energy conversion process within PV/T systems using numerical-based methods. In principle, several factors, such as fluid dynamics, thermodynamics, and radiative heat transfer within the system, provide a comprehensive understanding of the fluid behavior and heat distribution [98]. Generally, CFD models are computationally exhaustive but offer highly accurate results and capture complex characteristics within these systems [99]. Various studies have reported three-dimensional studies using different cooling mechanisms to simulate and model the PV/T performance under different parametric uncertainties, determining the conversion efficiency using CFD modeling techniques based on defined material properties and boundary conditions [93,100–106].

2.3.2. Analytical and empirical relation-based modeling technique

Analytical modeling techniques are also employed based on simplified mathematical expressions that resemble the behavior of PV/T systems. Although the models provide rapid computational insights into the system performance, they may not accurately retrieve all complexities

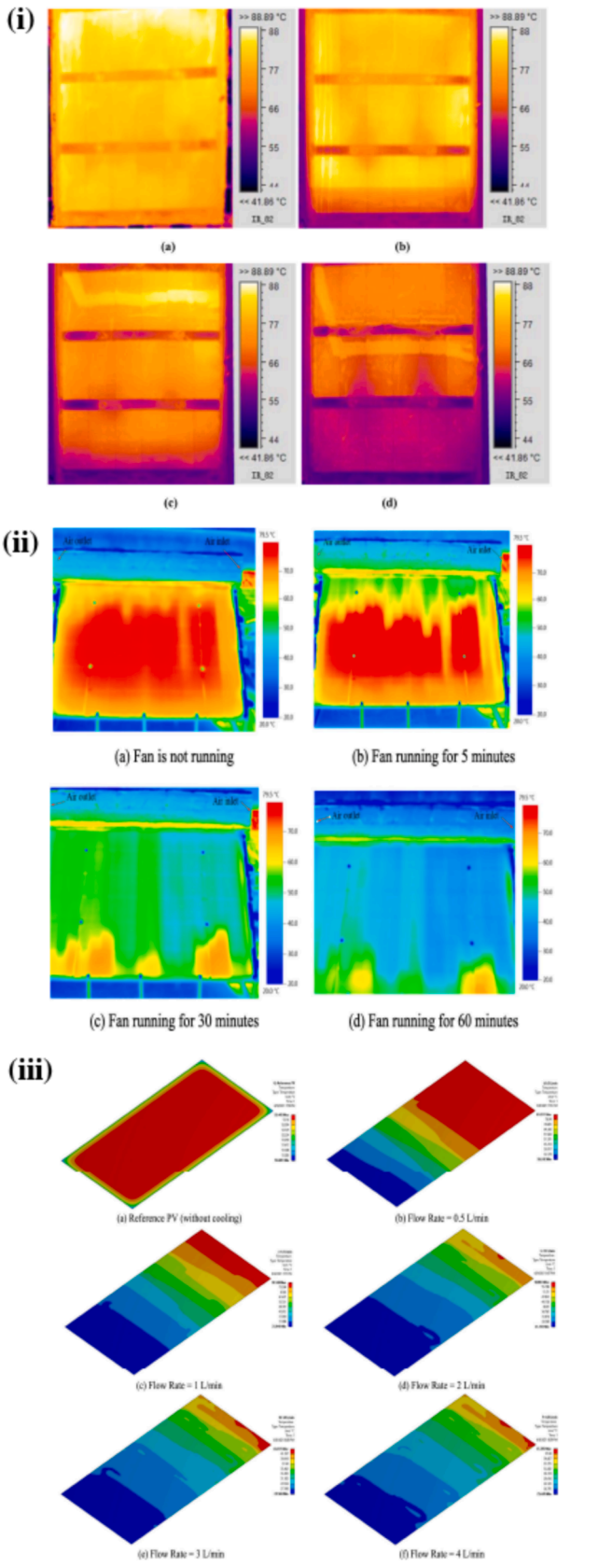


Fig. 3. Investigated heat patterns using different cooling methodologies (i) PCM material of thickness 1 cm, 2 cm, and 3 cm (Open Access. CC-BY) [87] (ii) Air cooled micro heat pipe array [88] (With Permission, License Number: 5813630919290) (iii) Water PV/T collectors under different flow rates [89] (With Permission, License Number: 5815230123025).

and interactions [107–109]. Moreover, empirical formula-based models are a different modeling technique developed for PV/T systems. Typically, empirical models are created based on retrieved experimental data from established PV/T systems. Empirical models estimate system performance based on the defined correlations. However, such modeling techniques may lack general employment beyond the specific experimental conditions or system specifications [110–115].

Consequently, prediction accuracy can be mitigated by developing semi-empirical models. The employment of semi-empirical models combines both analytical and empirical model aspects. Such aspects include fundamental physics-based mathematical models besides the experimental data retrieved to enhance prediction accuracy. Thereby, semi-empirical models consist of both complex and simple mathematical models, providing higher accuracy and efficient computation [116–118]. An example of a semi-empirical model developed is the modified single-diode model proposed by Gholami et al. [119]. The model predicts electrical behavior based on defined correlations between dust accumulation impact and various environmental conditions. Fig. 4 explains the semi-empirical model developed [119] to evaluate PV system performance under certain parametric conditions. Gholami et al. [120] also proposed five semi-empirical correlations to accurately predict PV cell temperature with different temperature coefficients based on experimental and analytical examination. The proposed semi-empirical correlation provides precision and accuracy to study the behavior of PV systems in harsh climatic conditions. In this notion, different semi-empirical models may be developed to accurately predict the performance of PV/T systems under specific parameters using real-time experimental results to validate the proposed model.

2.3.3. *Electrical equivalent circuit modelling (EECM)*

The PV cell depends on the variation of environmental parameters such as solar irradiance and operating temperature. Thus, a different classification of equivalent-circuit models has been developed to accurately imitate the electrical characteristics of given PV cells/modules [121,122]. Different equivalent circuit models have been classified as presented in Fig. 5, providing different modeling complexity and

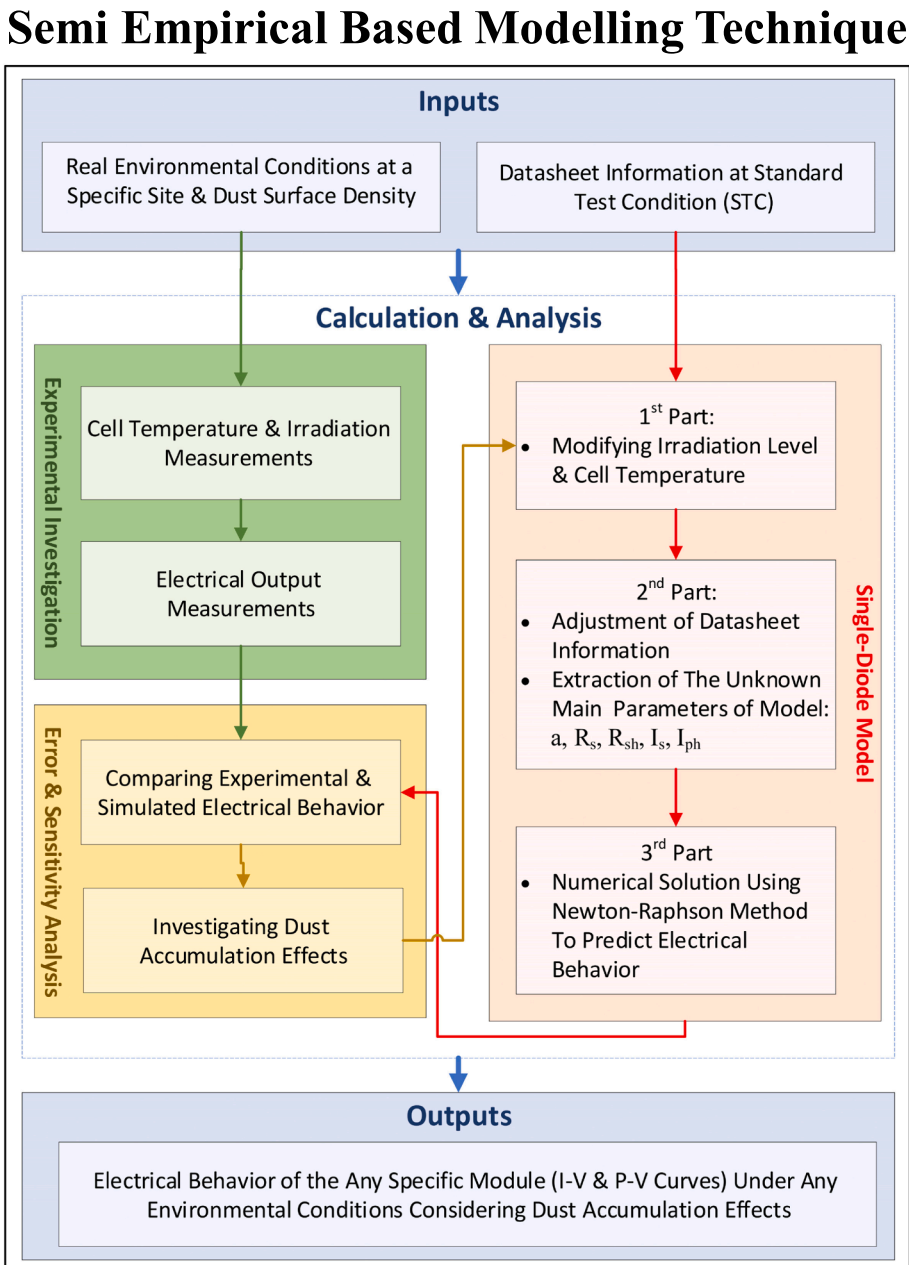


Fig. 4. Proposed Framework for Semi-Empirical Study proposed by [119] (With Permission, License Number: 5815230319481).

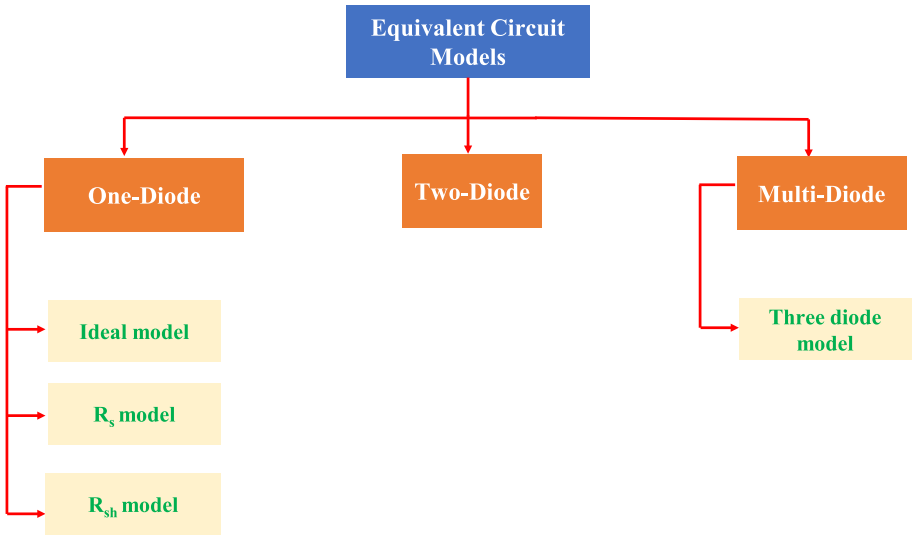


Fig. 5. Classification of PV cell equivalent circuit models.

accuracy rates in electrical characteristics imitating the experimental conditions.

In principle, the one-diode model establishes the fundamental configuration of a PV cell. The ideal configuration presents the PV cell without accounting for possible losses, classifying it as impractical for replicating the electrical characteristics of the PV cell in natural conditions [123]. The output current of this configuration is typically represented by the Eq. (8) as follows [124]:

$$I = I_{sol} - I_s\left(\exp\left[\frac{qV}{kT\gamma N_c} - 1\right]\right) \tag{8}$$

Where I_{sol} is the photo-generated current, I_s is the saturation current, V is the output voltage, q and k resemble the electron charge and Boltzmann constant, respectively. γ is the ideality factor, N_c resembles the number of cells connected in series, and T is the P-N junction temperature.

The ideal model was further developed to account for the possible losses in the PV cell by adding a series resistance R_s. The output current of this configuration is typically represented by Eq. (9) as follows:

$$I = I_{sol} - I_s\left(\exp\left[\frac{q}{kT\gamma N_c}(V + IR_s) - 1\right]\right) \tag{9}$$

The model was utilized and reported in several investigations, reporting that introducing a series resistance improves accuracy when extracting the electrical characteristics of a PV cell [125,126].

A highly popular configuration is adopted that introduces a parallel resistance named the shunt resistance. The shunt resistance is typically responsible for the leakage current across the PV cell [127]. The output current of this configuration is represented by Eq. (10) as follows [128]:

$$I = I_{sol} - I_s\left(\exp\left[\frac{q}{kT\gamma N_c}(V + IR_s) - 1\right]\right) - \frac{V + IR_s}{R_{sh}} \tag{10}$$

This configuration of the one-diode model has been widely implemented in various areas due to the accuracy of the model [129]. A comparison between the experimental and model-based electrical characteristics presented a 10 % error based on the proposed model [130]. Moreover, Lo Brano et al. [131] proposed the continuous improvement of this model by introducing a thermal coefficient to estimate the electrical characteristics accurately under temperature fluctuation. Additionally, optimization-based methods and machine learning significantly improve the model's accuracy [132,133].

The shunt resistance-based one-diode model provided efficient accuracy under climatic condition variations such as solar irradiance and operating temperature. However, there is a reduction in the model's performance during minimal changes in solar irradiance and operating temperature and not abrupt changes specifically. This dilemma was overcome by introducing a two-diode equivalent model [134]. The output current of this configuration resembles two diodes in parallel, as presented in Eq. (11) as reported by [135]:

$$I = I_{sol} - I_{s1}\left(\exp\left[\frac{q}{kT\gamma N_c}(V + IR_s) - 1\right]\right) - I_{s2}\left(\exp\left[\frac{q}{kT\gamma N_c}(V + IR_s) - 1\right]\right) - \frac{V + IR_s}{R_{sh}} \tag{11}$$

The two-diode model defined in Eq. (11) provides a complex modeling procedure due to seven unknown parameters. Essentially, these parameters are estimated through various numerical methods. Numerical methods have been proposed by employing artificial intelligence-based methods to tackle the issue of the two-diode model complexity [136]. Additionally, the employment of objective functions to estimate the optimum parameters for the model was discussed in [137]. However, numerical methods tend to have a higher computation time due to their dependency on iterations. Therefore, efficient techniques have been proposed to present high accuracy with a quick response rate, as presented in [138].

Finally, with the rapid development of approximation techniques, multi-diode models have been introduced to the previous configuration to provide improved electrical characteristics compared to the previous models [139]. The output current of such configuration can be expressed through Eq. (12):

$$I = I_{sol} - I_{s1}\left(\exp\left[\frac{q}{kT\gamma N_c}(V + IR_s) - 1\right]\right) \ldots - I_{sn}\left(\exp\left[\frac{q}{kT\gamma N_c}(V + IR_s) - 1\right]\right) - \frac{V + IR_s}{R_{sh}} \tag{12}$$

Involving multiple unknown parameters raises complex uncertainties that must be resolved for using such an electrical equivalent model. The complexity can be determined by introducing metaheuristic optimization techniques to optimize the equivalent unknown parameters, as presented in [140] for a three-diode equivalent circuit model. Moreover, it is mainly observed that in using a three-diode equivalent circuit model, the accuracy of predicted electrical characteristics is much higher than that of single- and double-diode configurations [141,142].

Although equivalent circuit models have been developed to predict

the electrical characteristics of PV modules, adopting such models is solely used to examine electrical characteristics. The single-diode model is prominently employed in various studies due to its modeling simplicity and parameter estimation [143–145]. In some instances, the employment of a double-diode equivalent model configuration was approached due to its superior performance and accurate prediction of electrical characteristics [146,147]. However, developing such electrical models lacks the accurate replication of electrical characteristics in PV/T systems, presenting certain limitations in their employment. An investigation by Tina et al. [148] was conducted to assess the impact of temperature on electrical characteristics based on different cell connections through a numerical study.

Moreover, Bechouat et al. [143] discussed the limitation of abrupt temperature on PV cells on electrical characteristics and overall electrical performance. Therefore, the study proposed a numerical-based model for PV/T systems through a 2D lookup table and the single-diode cell configuration to accurately simulate temperature impact on system performance. Moreover, reported investigations by [144,145] utilized single and double-diode configurations to estimate the effects of environmental conditions, particularly temperature, on the PV/T system's performance.

In this notion, the impact of operating temperature on PV systems is not simulated and predicted accurately with contemporary electrical-based models. Essentially, the development of models that provide a combination of thermal and electrical factors needs to be considered to observe the impact of temperature on the system's complete performance.

2.3.4. Machine learning and optimization modeling techniques

Artificial intelligence (AI) has grown exponentially in various areas of interest, benefiting research and development [149–152]. In particular, AI-based modeling techniques are advantageous in deriving proper relations between input and output parameters [153–155]. Additionally, incorporating neural networks and metaheuristic optimization can be employed to forecast the maximum power achieved from PV systems with high prediction accuracy [156–160]. The efficiency of photovoltaic-thermal systems is usually examined based on experimental investigations or computational tools such as CFD, as presented earlier [161,162], which consume a considerable amount of resources. In addition, various computational models have been theoretically analyzed based on mathematical expressions using various assumptions, decreasing the forecasting accuracy.

AI significantly improves photovoltaic-thermal (PV/T) modeling by understanding system input and output non-linear relationships [163]. They are simplified, quick, and effective in computing nonlinear relationships between input and output [164]. AI algorithms can be

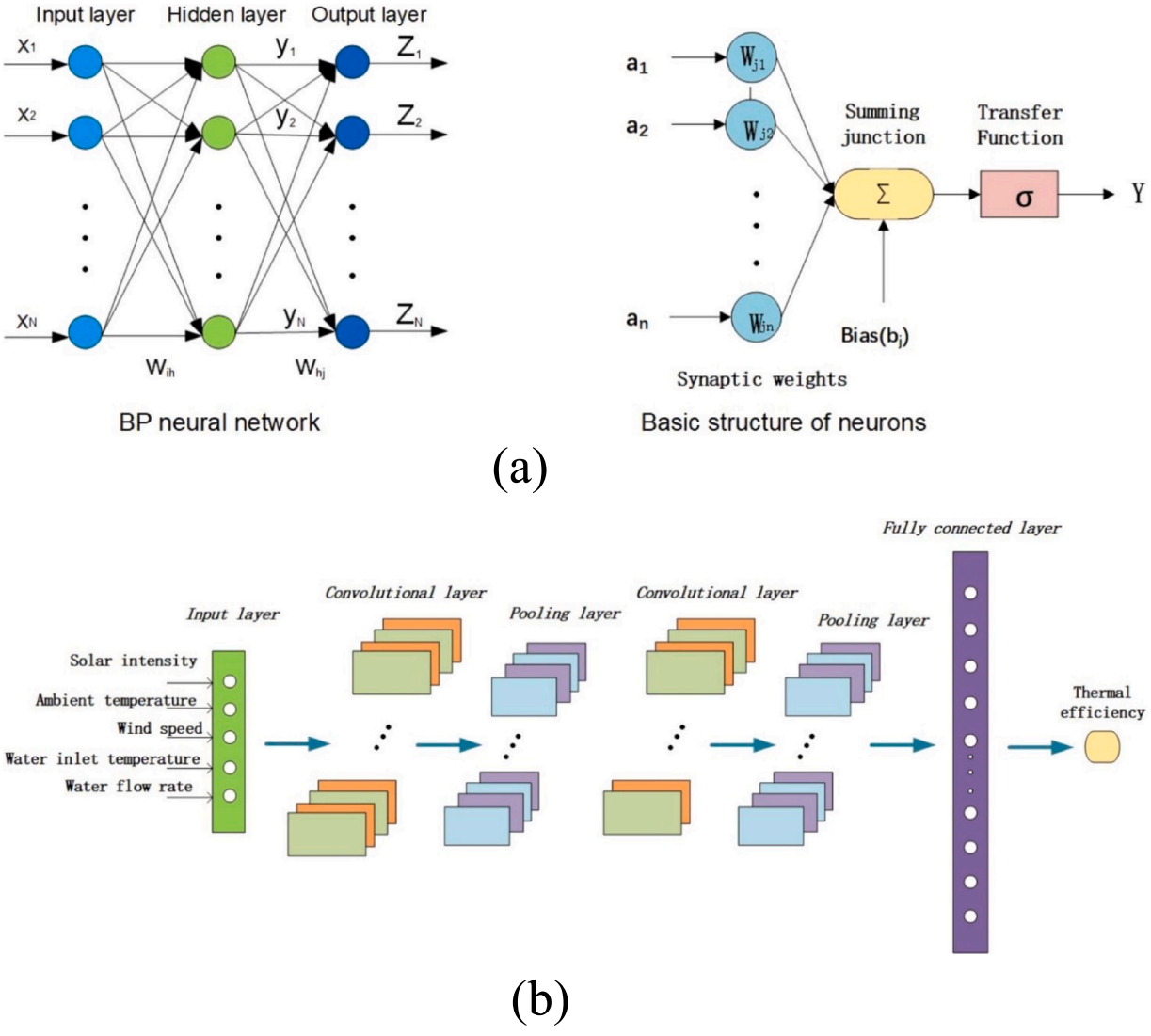


Fig. 6. Typical machine learning structure for PV/T system predictive models (a) Back Propagation (BP) neural network models [155] (b) Convolutional Neural Network (CNN) model [155] (With Permission, License Number: 5815230621063).

employed to develop predictive models that estimate and predict the performance of photovoltaic-thermal cooling systems [165]. In particular, examining historical weather parameters, system parameters, and energy generation data enables AI to be trained under different data patterns and accurately predict PV/T system operation in various conditions. A study presented by Wang et al. [166] developed a relationship model for accurate forecasting of annual exergy generation of an air-based cooling PV/T system. The model proposed an artificial neural network (ANN) and a genetic algorithm (GA) optimizer. The proposed model provided that the yearly exergy can be increased to 6000 kWh by increasing the channel width and length to 5 m and 10 m, respectively. Popular methods are used for performance prediction, such as Back Propagation neural networks (BP) and Convolutional Neural Networks (CNN), which have been widely used in the scientific literature for deriving input and output system parameter relationships [167,168]. BP is considered a benchmark in developing predictive models for comparing the performance of different systems under different uncertainties [169]. As demonstrated in Fig. 6(a), input variables are considered independent parameters crucial in defining the system's overall working predictive model [170]. Moreover, CNN is based on three stages: convolution, pooling, and fully connected layers, where an input layer consists of the independent parameters, and the convolution and pooling layers act as interstage layers [171,172]. The interstage layers can be increased depending on the desired depth and performance needed from the CNN, as demonstrated in Fig. 6(b).

AI-based tools have been further reported and investigated to develop predictive models, which assess the thermal performance of PV systems and predict their behavior based on trained datasets. The PV module temperature was predicted for building-integrated photovoltaic systems (BIPV) based on the employment of Grammatical and Differential Evolution methods, developing a tailored expression based on the popular Sandia model as reported by [173]. The developed model presented an error below 4 % during different climatic conditions, reducing the error by 11 % compared to the traditional Sandia model. The demonstration of a hybrid Artificial Neural Network and Particle Swarm Optimization (ANN-PSO) was reported to forecast the energy performance of the BIPV/T system [174]. Input parameters consisted of channel length, depth and width, and the air mass flow rate, which were used to evaluate the overall performance and determine a relationship between these parameters. In this notion, the proposed hybrid model was superior in its performance during the training and testing phase, providing minimal error in its prediction accuracy. Another investigation on machine learning techniques reported using Gaussian process regression (GPR) for day-ahead prediction to examine the performance of a PV/T system based on water and nanofluids as cooling fluids [175]. The study outcomes reported comparable data between forecasted and experimental data, accurately predicting PV/T electrical and thermal performance. Another study assessed the employment of various AI-based models to determine the one with the highest accuracy for predicting PV/T system electrical performance using nanofluids as a coolant [176]. The models that are employed to predict the electrical efficiency of PV/T systems using nanofluids coolants were adaptive network-based fuzzy inference systems (ANFIS) [177], multi-layer perceptron (MLP) [178], cascade feedforward (CFF) [179], radial basis function (RBF) [180], generalized regression (GR) neural networks [181], and least-squares support vector regression (LS-SVR) [182]. The study reported that ANFIS presented the highest prediction accuracy based on 263 experimental datasets reported from the scientific literature. Moreover, a differential evolution optimizer was employed to maximize the PV/T electrical efficiency generation and to determine the optimum condition and objective function. Additionally, researchers have sought to reduce the mathematical complexity provided by analytical models representing the PV/T system's performance to benefit from rapid processing time and high forecasting accuracy. ANN, ANFIS, and LS-SVR have been popularly employed and used for optimum forecasting of electrical and thermal efficiency of PV/T systems based on PCM and nanofluid coolants [183], while least square support vector machine (LV-SVM) also showed notable merits as compared to ANN in predicting the thermo-electrical efficiency of a hybrid PV/T system [184]. Additionally, metaheuristic optimization has been widely integrated within AI predictive models to maximize the conditions to achieve optimum electrical and thermal efficiencies and to maximize the overall performance of PV/T systems [185–191].

Additionally, recent literature has focused on implementing AI and machine learning tools to optimize the design and operation of building integrated photovoltaic-thermal (BIPV/T) systems. The combination of novel metaheuristic optimization algorithms and machine learning techniques enables to evaluate the performance of coolants such as water, air, and thermoelectric to forecast the efficiency and applicability of these systems both from the energy and economic perspective, proposing relatively high prediction accuracy [192–194]. Utilization of fuzzy logic based on the Auto-Regressive Moving Average (ARMAX) function was incorporated to study the design of a water-based PV/T collector. The fuzzy control system can comprehend the system's complexity by suitably designing the optimum water flow accordingly [195]. Additionally, the incorporation of phase change material and nanofluids have been optimized based on predictive modeling using different machine learning techniques such as neural networks, artificial neural networks, extreme gradient boosting (XGB), extra tree regression (ETR), and k-nearest-neighbor (KNN) regression models, developing accurate prediction models for the system output performance based on experimental datasets [196–198].

Typically, machine learning models are composed of multilayers that define variables for input and target data [199,200]. In principle, data such as volume fraction of base fluid, PV/T inlet temperature, and time are categorized as input variables. Target variables are considered the objective variables, comprising PV module temperature, the thermal difference between inlet–outlet, and electrical and thermal efficiencies. Data is typically classified into linearly separable data to extract its features. Such models comprise input variables that present different features in the presented dataset. Target features represent the output parameter that needs to be predicted and forecasted based on the optimization.

The input and target features are interconnected with hidden layers in the multilayer perceptron (MLP) architecture. Each hidden layer is defined by random weights (W) and bias (B), which are computed on iteration [201]. The net network input feature is calculated as follows:

$$Y_{in} = [X] \times \left[W^{T}\right] + B \tag{13}$$

The output features of the MLP networks are typically controlled using an activation function, and their weight is adjusted by applying various optimization techniques [202].

$$Y_{out} = \varphi[Y_{in}] \tag{14}$$

Where φ represents the activation function, the network error (E) can be computed as a difference between the target feature and the MLP output.

$$E = T - Y_{out} \tag{15}$$

The weight and bias can be updated through backpropagation. The backpropagation equation is typically presented as follows:

$$J\alpha(W, B) \tag{16}$$

$$\frac{DE}{DB} = 0 \tag{17}$$

$$\frac{DE}{DW} = 0 \tag{18}$$

The weights and updates are typically updated based on the minimum network error. The learning rate (α) is used for the testing purpose. Thereby, the mathematical relation for testing can be expressed as fol-

lows:

$$\text{Testing} = \psi\left[\text{test}_{\text{data}} \times \left(\text{W}_{\text{updated}} + \text{B}_{\text{updated}}\right)\right] \quad (19)$$

Finally, the efficiency of the proposed machine learning algorithms can be monitored and compared based on the mean square error (MSE) and mean absolute error (MAE) as follows:

$$\text{MAE} = \frac{1}{\text{N}} \sum_{\text{j}=1}^{\text{N}} \text{y}_{\text{j}} - \widehat{\text{y}}_{\text{j}} \quad (20)$$

$$\text{MSE} = \frac{1}{\text{N}} \sum_{\text{j}=1}^{\text{N}} \left(\text{y}_{\text{j}} - \widehat{\text{y}}_{\text{j}}\right)^2 \quad (21)$$

In this notion, the employment of AI-based tools assisted with optimization techniques provides a substantial enhancement in developing predictive models to forecast electrical and thermal efficiencies of PV/T systems and their respective coolants. Moreover, their employment would reduce computational complexity and add advanced modeling techniques, providing an accurate rate of forecasting and intelligent control based on set input parameters, hence resulting in optimizing system design and performance. However, AI-based tools require sufficient data for the training and testing phases to reach a desirable prediction accuracy with minimal error compared to the experimental data achieved.

Thereby, the classification of these modeling techniques provides a framework for different approaches to study and optimize the performance of photovoltaic-thermal systems to benefit from the system in terms of maximum energy efficiency, as presented in Table 1. The selection of modeling technique depends on the required objective of the study, the available data to be used, and the preferred level of computational accuracy and complexity to compare the impact of cooling on PV systems mathematically based on the experimental data derived.

Table 1
Modeling and Computation techniques classification for PV/T systems.

Technique	Summary
Computational Fluid Dynamics (CFD)	Numerical technique to simulate fluid flow and heat transfer in PV/T systems, providing a visual representation of the system's performance to improve the system's energy efficiency under several cooling configurations.
Analytical Modelling	Simplified modeling technique is dependent on mathematical computations for electrical and thermal characteristics estimation without the need for simulations.
Empirical Modelling	A simplified modeling technique is based on establishing statistical relationships between input parameters and the system's output performance based on experimental data.
Semi-Empirical Modelling	Modeling technique combining elements of empirical and physical modeling techniques. Typically, empirical data is employed along with fundamental physical properties to capture the behavior of PV/T systems with higher accuracy.
Electrical Equivalent Circuit Modelling (EECM)	EECM modeling techniques purely capture the electrical characteristics under varying climatic conditions. Such a modeling technique has limited use since it is restricted to electrical characteristics prediction without simulating coolant impact.
AI-based Modelling	AI-based modeling techniques are implemented to develop predictive models for PV/T systems based on datasets that consist of historical energy and weather patterns. Under training and testing phases, AI can develop tailored relationships and data prediction patterns for accurate forecasting of energy generation under cooling impact. Additionally, AI-based tools are assisted with optimization techniques to maximize the boundary conditions for energy generation.

2.4. Adaptive cooling frameworks for smart and optimized PV/T cooling

The scientific literature has explored the adoption of PV/T systems and the employment of active and passive cooling methodologies rigorously, highlighting the various technologies introduced to enhance the overall electrical and thermal performance of PV energy systems. Multiple review articles have reported the use of cooling methodologies through air-cooling, front-surface cooling, thermoelectric generators (TEGs), Phase change material (PCM), and nanoparticles, highlighting the notable improvement of providing a cooling method to lower operating temperature [64,203–208] Moreover, a rich comparison of modeling and computation techniques have been highlighted and reviewed, providing understanding over the physical dynamics of PV/T system both in terms of electrical and thermal properties, as a means of PV system performance enhancement and developing predictive models for energy forecasting under cooling methods [209–211].

Prominently. The deployment of water as a working base fluid has been popularly used within the scope of scientific literature as a means of enhancing the electrical power output due to its feasible employment and availability as compared to other fluids, providing promising potential [212–214]. Additionally, the deployment of water as a working base fluid is cost-friendly compared to other possible fluids and materials that are difficult to apply on a large scale [215,216]. Besides, water serves a dual purpose for both cooling and cleaning applications to boost the performance of PV energy systems [217–223]. Thereby, the selection of water as a working base fluid presents a suitable solution for PV/T system enhancement [224].

However, the wide range of PV/T systems and applications discussed mainly highlighted the deployment of such cooling techniques without considering the provision of robust and adaptive control techniques to facilitate effective temperature and dust monitoring [225]. The essential adaptability of such systems needs to be encouraged to effectively be integrated in real-time with PV systems to sustain their performance in the long term [226–228]. Notably, the adoption of such schemes provides a method of reducing the degradation effect by maintaining a monitoring scheme through defined temperature boundaries, as presented in Fig. 7.

In this notion, the prospect of providing adaptive and robust control cooling frameworks promotes the adaptability of PV systems within various climatic conditions and harnesses optimal maximum power points while maintaining an optimal operating temperature that is not to be exceeded. Limited investigations have reported the utilization of adaptive and robust control active cooling frameworks. Such methodologies adopt various microcontroller technologies, integrated sensors, and electronic components to facilitate feasible automation and smart active cooling systems for PV energy systems, enabling anticipation of operating temperature and correspondingly maximizing the PV system's maximum power output.

3. Adaptive and controlled cooling frameworks

This section explores various reported state-of-the-art adopting automated and innovative frameworks for active cooling processes on PV energy systems. Furthermore, the derived results from the highlighted state-of-the-art systems provide a potential prospect for adaptability with PV energy systems.

3.1. Adaptive control schemes and state-of-the-art

Providing an adaptive scheme to sustain the PV energy system's long-term energy production is critical to maintaining its stability as well as integration within power systems. Optimization of PV operating temperature is essential to maximizing electrical parameters such as maximum operating voltage (V_{mpp}) and maximum power point (P_{mpp}). Therefore, a controlled and adaptive cooling methodology is essential to tackle and reduce abnormal boundaries in operating temperature and

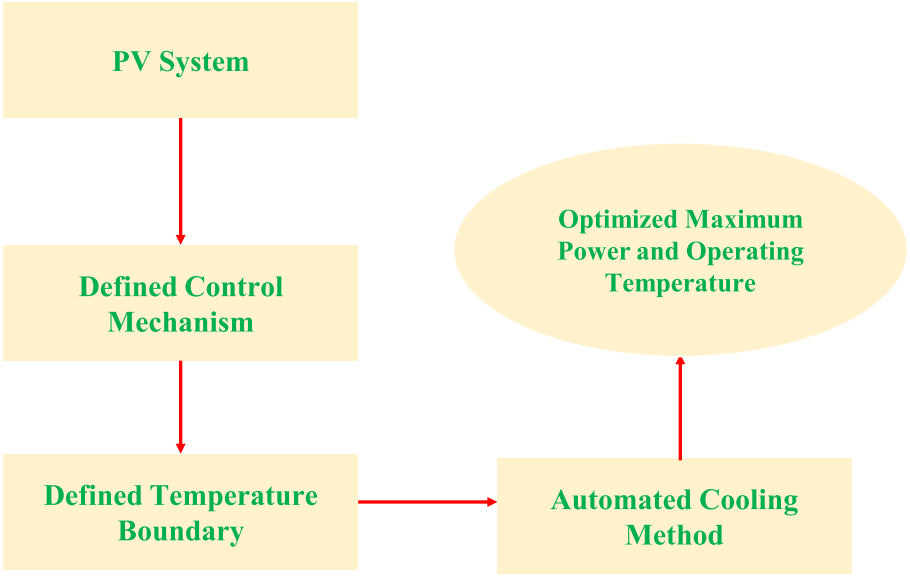


Fig. 7. Analogy for Automated Cooling Methods.

cope with the possible accumulation of dust deposition.

In principle, microcontroller technologies and sensors have been integrated within PV systems as a means of providing flexible operation and access to their performance. In particular, PV monitoring systems have recently been adapted through various sensors and data transmission to facilitate active monitoring of the system's performance for their predictive maintenance [229–231]. A wide range of microcontroller technologies are used. However, NodeMCU microcontrollers and Arduino microcontrollers are popularly implemented for their flexibility in data transmission and integration with sensors, providing wireless interlinking with established wireless networks. Both technologies are feasible in their integration, making them popular choices for developing monitoring systems for PV systems. Additionally, the integration of internet connectivity is viable and compatible with both technologies, enabling the design of a friendly user interface for interaction.

However, limited studies have explored the process of automated cooling frameworks rather than the manual process to provide a smarter prospect in PV cooling systems and propose control strategies based on microcontroller technology and integrated sensors. The reported studies in the scientific literature have adopted temperature boundaries to trigger the cooling mechanism automatically rather than adopting manual cooling operation, providing a monitored procedure to serve the consumer and system effectively in terms of dust cleaning and temperature reduction to reduce PV systems' long-term degradation [232]. A wide range of proposed technologies have been employed to successfully facilitate automated methods of detection for PV operating temperature, as presented in Fig. 8.

Moreover, water as a working base fluid is prominently employed for cooling for its feasible employment, flexible system suitability, excellent heat extraction for use in preheating applications, and dust cleaning methods. Therefore, multiple researchers have sought to tackle the dilemma of increased PV module operating temperature by introducing adaptive and controlled techniques for base-fluid circulation to enhance the overall efficiency of the PV system. M. Kamarudin et al. [233]

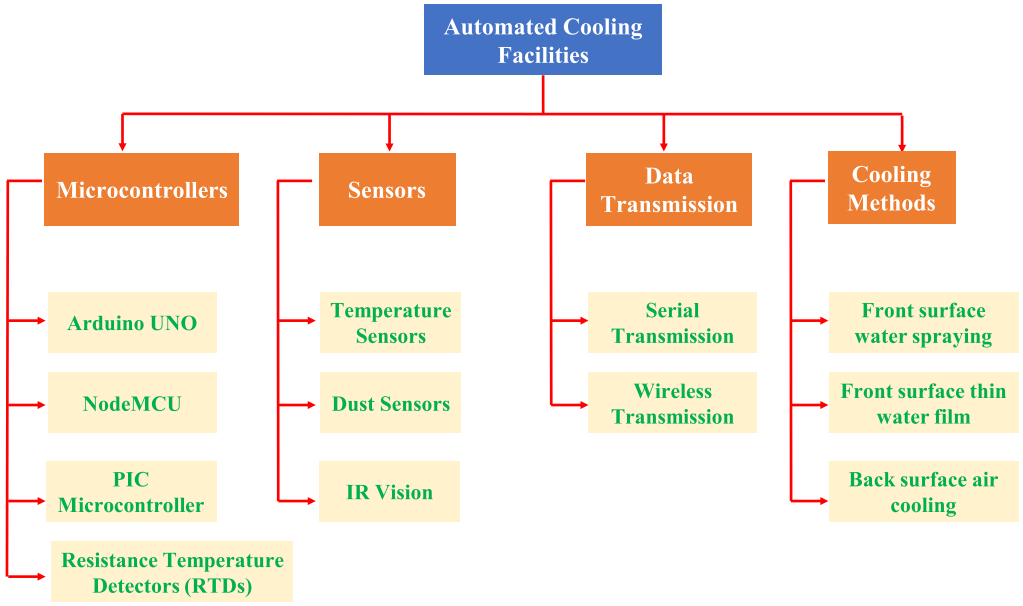


Fig. 8. Technologies for automated and adaptive cooling frameworks reported from the scientific literature.

proposed the maximum power harvesting for PV systems by introducing an active front-surface cooling methodology using an Internet of Things (IoT) facility. The reported study proposed a solution to cope with the degradation of the PV system's performance due to rising temperature limits. The deployed state-of-the-art provided a conventional Arduino Uno microcontroller to facilitate the automatic detection of temperature and active cooling process for a small-scale 20 W PV energy system. Furthermore, a temperature limit of 45 °C is defined and programmed to trigger a 12 V DC water pump for active heat extraction. A. Taqwa et al. [234] further explored and reported the employment of an automated active cooling process to optimize PV operating temperature and maximize maximum energy yielded based on front-surface cooling methodology. The experimental study adopted a control circuit to adopt front-surface cooling in the environmental conditions of Palembang, Indonesia, to enhance the performance of a 100 W PV state-of-the-art. Additionally, integrated temperature sensors adapted to control the cooling procedure based on a 36 °C reference temperature limit.

O.T. Laseinde [235] reported the employment of resistance temperature detectors (RTDs) to measure operating temperature and coupled them with a conventional Arduino Uno Microcontroller to facilitate the active cooling process. The experimental study reported the adoption of water as a base fluid for active cooling within the climatic conditions of South Africa. The automated cooling process adopts comparative circuits to estimate the voltage difference between the input RTDs, hence facilitating the switching mechanism for cooling effectively under hot tropical regions.

More advanced and sophisticated techniques have been reported using Arduino Uno and IoT to optimize PV/T system operation. A.R. Prasad et al. reported the optimization of PV module performance through an automatic cooling mechanism [236]. The experimental study investigated the employment of a conventional Arduino Uno microcontroller as an economical approach as compared to commercial-based controllers. The study examined the performance of a 60 W PV system in the climatic conditions of India based on active cooling. Additionally, a mini-DC water pump facilitates the automatic cooling methodology based on temperature sensors and voltage and current sensors to record electrical parameters.

M.M. Farag et al. [63] experimentally investigated an adaptive front-surface cooling framework to tackle the impact of harsh climatic conditions on PV energy systems during the arid summer of Sharjah, United Arab Emirates. The study was conducted on a small-scale 55 W PV energy system to assess the impact of automated front-surface cooling during extreme PV operating temperatures. A control circuit is deployed by implementing an Arduino Uno microcontroller and MAX6675 thermocouple temperature sensor to facilitate a 55 °C reference temperature limit for the automated cooling process through a 12 V DC water pump.

Furthermore, A. Benato et al. [237] have proposed an active front-surface cooling methodology based on deploying automated on/off modes of operation for PV surface spraying. The study investigated the deployment of different nozzle configurations such as strip, 90°, and 180° to decide the optimum geometrical configuration for maximum temperature reduction. Additionally, N. Nader et al. [238] investigated the deployment of an automated cleaning and cooling mechanism employed for a 320 W PV energy system in the climatic conditions of Al-Khobar, Saudi Arabia. The experimental study evaluated the active cooling process employed based on its improved electrical efficiency using water as a base fluid.

Finally, S. Paul et al. [239] employed a smart-cooling mechanism based on microcontroller technology to decrease the abrupt temperature increment in PV energy systems. A reference temperature limit is set at 25 °C to start the automated cooling process. The experimental study consists of a 20 W capacity PV module and DC sensors to monitor electrical parameters and detect temperature to facilitate the active cooling process. Furthermore, L. Kumar et al. [240] experimentally investigated the loss experienced in the electrical conversion efficiency as a result of the abrupt reference temperature limits, which provides a severe impact on the performance of the PV energy system. The experiment was conducted through a configurated LM35 temperature sensor with a reference temperature of 45 °C for automatic waste heat extraction using water as a working base fluid, as presented in Fig. 9. The analog temperature signal is fed to configured PIC microcontroller to operate a MOSFET switch through pulse width modulation (PWM) to activate an automated spraying mechanism to enhance the PV system's performance.

In addition, I. Cho et al. [241] demonstrated the use of a back-surface active cooling process using air as a convection medium through a cooling fan methodology. The study investigated the use of four cooling fans mounted on the back surface of a 30 W PV module to reduce the operating temperature effectively. The experimental investigation was conducted in the climatic conditions of Busan, Korea, with a defined 40° C reference temperature limit to automatically run four cooling fans of 360mW electrical capacity as presented in Fig. 10(a). Finally, Kusuma et al. [242] proposed and investigated a cooling method based on air cooling methodology using a DC fan, heatsink, and Peltier as presented in Fig. 10(b). By adopting Arduino Uno as a control unit, a defined temperature limit is set at 40 °C to facilitate cooling in Indonesia's climatic conditions. The experimental investigation was conducted at 30-minute intervals to determine the variability of environmental conditions.

Furthermore, an adaptive cooling model was developed to optimize the performance of a 2.88 kW PV system based on defined temperature thresholds [243]. The model was developed in a MATLAB environment utilizing real-time experimental data to facilitate adaptive front-surface cooling using water as a coolant under different weather conditions. A real-time dynamic model was developed and reported by [244], utilizing an artificial dynamic neural network (ADNN) using active fan cooling. The developed model is trained in five different cooling patterns to adapt to weather and environmental conditions optimally and is utilized for efficient cooling.

In this notion, the previously discussed studies elaborated on adaptive schemes to facilitate automated cooling, optimize PV operating temperature, and harness the optimum maximum power generation. As a result, the upcoming subsection elaborates on the reported results that signify the merits achieved through adaptive cooling frameworks for overall PV system performance enhancement.

3.2. Results from examined adaptive cooling frameworks

Typically, the discussed state-of-the-art reported results in terms of temperature and electrical parameters correspondingly highlight the overall improvement of the PV energy system through an adaptive and controlled cooling framework. Such adaptive frameworks provide a robust and controlled procedure to optimize operating temperature and harness optimized maximum power generation.

In [233], the proposed adaptive front-surface cooling methodology integrated with the small-scale PV provided prospective merits in maximum power point and electrical efficiency enhancement. The observed average power reported with cooling was 18.1 W compared to 14.97 W without cooling, highlighting the significant reduction in temperature. Furthermore, the enhancement in electrical efficiency was reported at 10.42 % with cooling compared to 8.64 % without cooling. In addition, the experimental investigation reported in [234] provided promising merits using a reference temperature limit of 36 °C for the automated cooling process. A reported increment in average output power difference of 2.9 W was reported with cooling, providing an average temperature difference of 9.9 °C. Moreover, an average increment of 0.4 % in electrical efficiency is observed, resulting in overall performance enhancement during the period of automated cooling.

Furthermore, the study conducted in [235] provided an overall increment of 16.65 % by adopting RTDs and comparator circuits to control the water-cooling system based on a micro-controlled unit. The reported power with and without cooling coincided with 6.26 W and

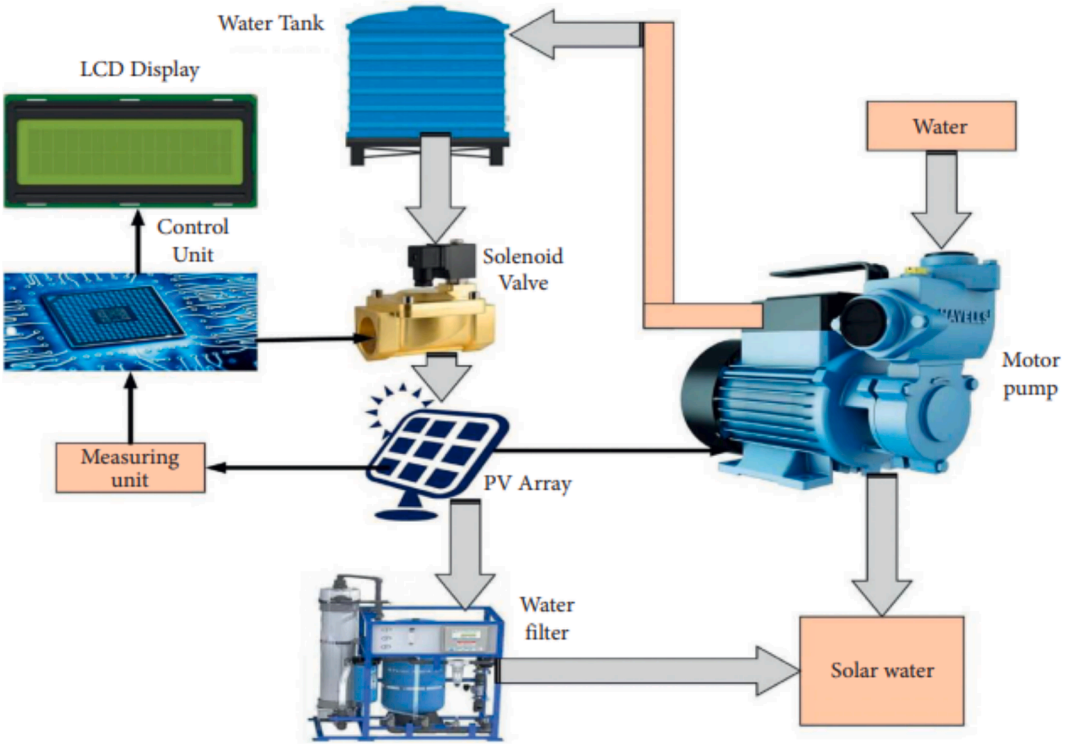


Fig. 9. Smart Spraying mechanism as demonstrated by [240] (Open Access, CC-BY).

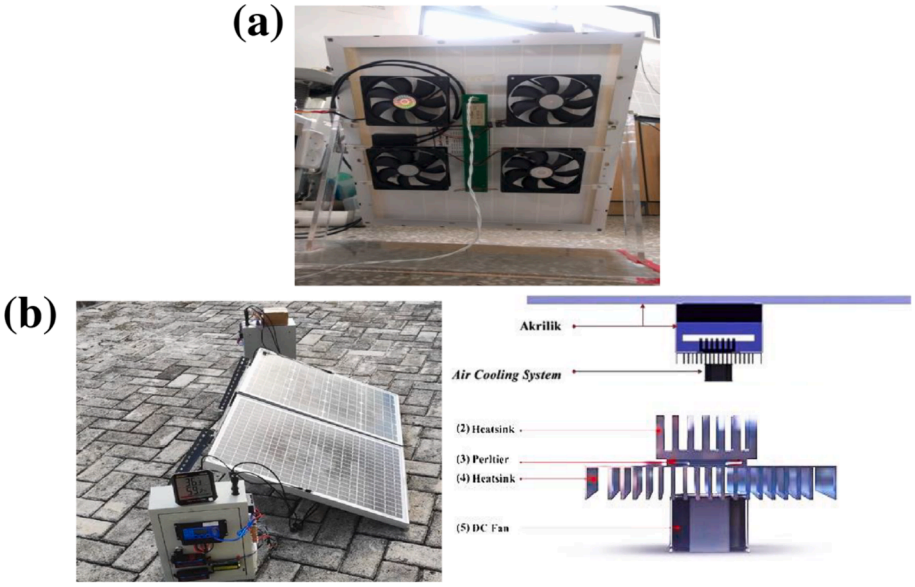


Fig. 10. Automated cooling using fans (a) [241] (Open Access, CC-BY-3.0) and (b) Automated cooling using fan and heat sink [242] (Open Access, CC-BY-SA).

5.37 W, respectively. Additionally, promising merits were reported in [236] for an automated and controlled rooftop PV system using front-surface water sprinkling. The study reported an enhanced electrical efficiency by 17 % through adopting an automated front-surface cooling methodology, with a reduction of 16 °C. The study highlighted the significance of implementing such a methodology using a conventional Arduino Uno Microcontroller due to its feasible integration with various modules and minor cost.

Moreover, the experimental study reported in [63] investigated adopting an adaptive cooling framework under harsh and arid summer conditions. An overall reduction in temperature is observed, reaching up to 17.3 °C, providing an extracted heat of 15.41 °C in terms of thermal heat as reported in the study based on IR imaging, comparing both harsh and moderate temperature distribution. Furthermore, the on/off mode-based PV spraying cooling system reported in [237] improved up to 13.27 % in electrical efficiency due to a maximum reduction in operating temperature up to 24 °C. Additionally, an increment of 33.43 W is observed per module, providing promising merits through active front-surface cooling. The experimental study reported in Khobar, Saudi Arabia, in [238] resulted in promising results within harsh regions through automated cooling and cleaning methodology. The electrical efficiency of the PV modules was reported to be from 10 % to 15 %

without cooling at an operating temperature of 42 °C.

The automated cooling process provided an overall decrement in temperature, reaching 20 °C, and an overall increase in electrical efficiency, which was reported as 7 %. In this notion, the maximum power and efficiency increased by 31 % and 32 %, respectively. An automated spraying procedure was proposed in [240] using a PIC microcontroller to adapt and operate the MOSFET switch for adaptive water spraying circulation. The study reported an enhancement in electrical efficiency of 0.5 % on average and an enhancement in power production. The mean temperature reduction reached 10 °C due to the intelligent water spraying procedure.

Back Surface cooling was introduced in [241] using four automated DC fans, tackling the temperature increment above 40 °C. A reported increment in electrical efficiency and maximum power was 4.7 % and 1.4 W, respectively. Additionally, back surface cooling using air was proposed by [242], setting a temperature limit of 40 °C for the controlled cooling procedure. The study reported a decrement of 14.5 % in operating temperature because of the cooling method. Moreover, an increment in average power output is observed, where 25.37 W and 23.82 W are achieved with and without cooling, respectively. Finally, a smart and automated cooling system was proposed using microcontroller technology by [239] to optimize PV operating temperature and maximum power generation. An increment between 1–2 W was observed through the cooling process, with a noteworthy reduction in operating temperature.

The developed adaptive cooling model reported in [243] presented a significant decrease of 25.36 % during September, which is considered one of the peak months in terms of module temperature. As a result, the adaptive cooling methodology significantly enhanced the electrical parameters such as maximum power output, maximum voltage, and electrical efficiency by 8.79 %, 6.39 %, and 6.58 %, respectively, based on a defined temperature threshold of 55 °C. The proposed real-time dynamic neural network model also presented robustness in predicting the optimal cooling pattern for 5 cooling fans [244]. The model provided an accuracy of 97 %, with a mean square error (MAE) of 0.02.

Overall, the experimental merits derived from the adopted frameworks and discussed established state-of-the-art have provided noteworthy enhancement to PV modules, mainly using water as a working base fluid, maximizing their performance, and sustaining their lifetime. Additionally, adopting automated and predictive cooling procedures for PV systems provides less variation in power generation in regions where solar irradiance exposure is rich and abundant, with significant exposure to IR radiations that contribute to the excessive heat on PV systems. Table 2 summarizes the state-of-the-art reporting adaptive and smart-monitored cooling frameworks.

4. Challenges and recommendations

The previous discussion highlighted the incorporation of optimization techniques such as traditional modeling through empirical modeling, electrical models, and computational fluid dynamics. Modeling techniques are well established and discussed within the

Table 2
Summary of established state-of-the-art reporting adaptive and smart-monitored cooling frameworks.

Ref	Location	System Scale	Framework Facility	Cooling Methodology	Temperature Reduction (°C)	Electrical Efficiency (%)		Average Power (W)	
						Reference	*Cooled*	*Reference*	*Cooled*
[239]	Bangladesh	20 W	Arduino UNO Microcontroller and remote monitoring application for automated cooling	DC water pump for front-surface circulation at 25 °C	Reduction of 5 °C	NA		16.95 W	19.55 W
[233]	Malaysia	20 W	Arduino UNO Microcontroller integration and Sensors for automated temperature detection with IoT facility based on a graphical user interface	Continuous Front-surface cooling using water at 45 °C	Maintained below 45 °C	8.64 %	10.42 %	14.97 W	18.1 W
[234]	Indonesia	100 W	Integrated Sensors Coupled with a Control Circuit and measurement tools	Continuous Front-surface cooling using water at 36 °C	Reduction of 9.9 °C	3 %	3.4 %	19 W	21.9 W
[235]	South Africa	17 W	Temperature detection based on Resistance Temperature Detectors (RTDs) coupled with comparators and Arduino UNO microcontroller technology	Water Spraying mechanism on PV module surface at 25 °C	NA	Overall Increment of 16.65 %		5.37 W	6.26 W
[236]	India	60 W	Integrated Arduino UNO microcontroller and module sensors for automated cooling process coupled with Data logging facility	Continuous Front-surface cooling using water at 34 °C	Reduction of 16 °C	Overall Increment of 17 %		48 W	53 W
[63]	UAE	55 W	Integrated Arduino UNO Microcontroller and Thermocouple sensors for temperature detection coupled with a control circuit for an automated cooling process under harsh climate	Continuous Thin-film front-surface cooling using water at 55 °C	Reduction of 17.3 °C	11 %	12.49 %	35.42 W	40.21 W
[237]	Italy	230 W	Adoption of on/off modes with different angles for water spraying nozzles	Continuous Front-surface cooling using water at 25 °C	Reduction of 24 °C	11.18 %	13.27 %	178.9 W	212.3 W
[238]	Saudi Arabia	320 W	Automated cooling and self-cleaning mechanism	Water Sprinkling system controlled by Valve	Reduction of 20 °C	Electrical Efficiency increased by 31 %		Maximum power increased by 32 %	
[240]	India	36 W	PIC-based microcontroller configured with LM35 temperature sensor for the active operation of a MOSFET switch	Front-Surface water spraying unit at 45 °C	Reduction of 10 °C	Average Increment of 0.5 % with cooling		Maximum of 34 W power increment	
[241]	Korea	30 W	Microcontroller Integrated with Sensors and DC Fans for automated cooling	Back-surface cooling via air at 40 °C	Reduction of 10 °C	Electrical Efficiency increased by 4.7 %		Maximum power increased by 1.40 W	
[242]	Indonesia	80 W	Integrated Arduino UNO microcontroller with sensors and DC fan for automated cooling	Back-surface cooling via air at 40 °C	Reduction of 6 °C	NA		Maximum power increased by 1.55 W	
[243]	UAE	2880 W	Developed Adaptive model on MATLAB for controlled cooling under different weather conditions	Front-surface cooling via water at 55 °C	Maximum reduction of 25.36 %	Electrical Efficiency increased by 6.58 %,		Maximum power increased by 8.79 %	
[244]	Jordan	100 W	Experimental State-of-the-art comprising of 5 cooling fans for trained ADNN	Back-surface cooling via air at 40 °C	Maximum reduction of 27.5 %	Electrical Efficiency increased by 42.85 %		Maximum power reached 90 W	

literature as a means of PV/T optimization. Still, implementing adaptive cooling methodologies is at an early stage of deployment, presenting several potential challenges and limitations. The following challenges and recommendations provide a roadmap for future investigations in improving adaptive cooling frameworks:

- The deployment of adaptive cooling methods within PV/T systems introduces additional complexity due to the required electronic components, such as sensors, actuators, and control units, potentially increasing both cost and maintenance. As a recommendation, the design of adaptive PV/T systems would essentially need careful assessment and selection of components, regular maintenance, and redundant systems should be implemented to minimize downtime and enhance system reliability.
- The energy that electronic components consume may offset the power generated. As a recommendation, the conduction of a cost-benefit analysis to assess the economic feasibility of adaptive cooling frameworks within PV/T systems. The optimization of energy consumption by utilizing energy-efficient components and controlling the cooling process based on real-time environmental conditions and PV system performance [245].
- Adaptive cooling frameworks using base fluids such as water must be evaluated for their effectiveness in harnessing thermal energy and waste heat. PV/T systems should optimize both electrical and thermal energy extraction to improve overall energy efficiency. It is recommended that thermal heat extraction be assessed in future investigations and that waste heat be utilized for preheating. This reduces the use of boilers and lowers electricity costs, enhancing overall system efficiency.
- The previously discussed studies are mainly focused on PV modules rather than large-scale practical PV/T applications. Investigating adaptive cooling frameworks on large-scale systems, such as grid-connected or stand-alone PV/T systems, is essential. This enables the evaluation of these frameworks to maximize the power harvested throughout the year, limiting temperature and dust accumulation and ensuring clean energy generation.
- The assessment of cooling facilities and compatibility requirements needs to be investigated based on the specific PV technology and cooling method. Therefore, optimizing system design and compatibility is essential [246].
- Advanced monitoring techniques evaluate the PV/T performance in environmental conditions and adjust cooling regulations accordingly. Therefore, robust monitoring methods are needed to precisely measure the module temperature, performance, and environmental conditions. Thus, advanced microcontrollers are investigated to test the robustness and response time to abnormal module temperatures and dust accumulation. Moreover, infrared imaging and robot vision can localize hotspots and avoid defects [247]. As previously discussed, using advanced models and machine learning techniques would optimize adaptive cooling methods based on real-time data, enabling user-friendly systems and sustaining PV system longevity [175,248].

5. Future work

For future work towards advancing PV/T systems, research should focus on:

- Enhance the precision and accuracy of mathematical modeling and simulations to predict PV/T system behavior under varying weather conditions. A primary focus on optimizing thermal and electrical dynamics would allow for identifying effective cooling strategies.
- Leverage AI and ML techniques to study large datasets for pattern recognition and predictive modeling. Therefore, developing intelligent control algorithms that adjust cooling mechanisms in real-time based on performance data and weather conditions.
- Further exploration on the integration of IoT for real-time control, utilizing sensors and wireless communication to automate and adapt cooling processes. Implementing adaptive frameworks would respond dynamically to abrupt changes in module temperature and other influencing factors, minimizing human intervention and enhancing PV/T system reliability.
- Exploration of large-scale system modeling for PV/T system performance, especially in harsh and remote areas where manual operation is challenging.
- Temperature prediction throughout the year will allow the optimization of electrical and thermal parameters, preventing abrupt changes in module temperature.

6. Conclusion

This review highlighted photovoltaic-thermal (PV/T) systems as a solution to increase PV systems' longevity through waste heat extraction. Cooling methodologies that improve the PV system, such as air-based cooling, water-based cooling, PCM, or PCM with nanofluids, were reviewed, showing promising potential in electrical efficiency and thermal efficiency enhancement. However, the essential need to optimize PV/T systems is necessary to harness maximum potential energy and minimize their cost during implementation. This review explored the optimization of PV/T systems operation from a modeling and experimental standpoint, highlighting different strategies for implementing PV/T systems. Various modern modeling techniques, such as AI-based, electrical-based, empirical, semi-empirical, CFD, and analytical modeling, have been reviewed. All modeling procedures stood out for their suitability and applicability in the PV/T system's design and process optimization, showing potential in improving experimental development.

Moreover, optimizing the experimental techniques for PV/T systems was typically highlighted through the proposition of self-automated and temperature-regulated systems. The review has highlighted and reported various investigations that incorporated self-automated techniques and advanced technologies for sustaining the performance of PV/T systems in the long run. Different technology classifications have been done to present the performance of electrical and thermal energy extraction, showing reduced costs and devices needed. Finally, this review paves the room for future research and development in proposing more robust controlled methodologies, intending to employ different control techniques and modeling strategies, incorporation of multiple microcontroller technologies, and assessment of real-time response to environmental conditions, which can be successfully integrated into long-term and self-sustained PV/T systems.

CRediT authorship contribution statement

Mena Maurice Farag: Writing – review & editing, Writing – original draft, Visualization, Investigation, Conceptualization. **Abdul-Kadir Hamid:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Maryam Nooman AlMallahi:** Writing – review & editing, Writing – original draft, Conceptualization, Visualization. **Mahmoud Elgendi:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] Aziz N, Hossain B, Lamb L. The effectiveness of environmental protection policies on greenhouse gas emissions. J Clean Prod 2024;450:141868. https://doi.org/10.1016/j.jclepro.2024.141868.

[2] Ahmadi A, El Haj AM, Jamali DH, Kumar R, Li ZX, Salameh T, et al. Applications of geothermal organic Rankine Cycle for electricity production. J Clean Prod 2020;274:122950. https://doi.org/10.1016/j.jclepro.2020.122950.

[3] Nooman AlMallahi M, Al Swailmeen Y, Ali Abdelkareem M, Ghani Olabi A, Elgendi M. A path to sustainable development goals: A case study on the thirteen largest photovoltaic power plants. Energy Convers Manag X 2024;22:100553. https://doi.org/10.1016/j.ecmx.2024.100553.

[4] Yousef BAA, Obaideen K, AlMallahi MN, Alajmi N, Radwan A, Al-Shihabi S, et al. On the contribution of concentrated solar power (CSP) to the sustainable development goals (SDGs): A bibliometric analysis. Energy Strateg Rev 2024;52:101356. https://doi.org/10.1016/j.esr.2024.101356.

[5] Tan JD, Chang CCW, Bhuiyan MAS, Minhad KN, Ali K. Advancements of wind energy conversion systems for low-wind urban environments: A review. Energy Rep 2022;8:3406–14. https://doi.org/10.1016/j.egyr.2022.02.153.

[6] Chen P, Wu D. A review of hybrid wave-tidal energy conversion technology. Ocean Eng 2024;303:117684. https://doi.org/10.1016/j.oceaneng.2024.117684.

[7] Din MI, Ameen S, Hussain Z, Khalid R. Advancements, challenges, and pathways to sustainable biofuel production. Chem Eng Technol 2024;47:614–23. https://doi.org/10.1002/ceat.202300263.

[8] Haj Assad M El, Kumar L, AlMallahi M, Rejeb O, Elgendi M, Farag MM, et al. Direct and indirect uses of geothermal energy. In; 50th Int. Conf. Comput. Ind. Eng. Sustain. Digit. Transform. CIE 2023, 2023, p. 1595–604.

[9] Kumar D, Mathur HD, Bhanot S, Bansal RC. Modeling and frequency control of community micro-grids under stochastic solar and wind sources. Eng Sci Technol an Int J 2020;23:1084–99. https://doi.org/10.1016/j.jestch.2020.02.005.

[10] Farag MM, Patel N, Hamid A-K, Adam AA, Bansal RC, Bettayeb M, et al. An optimized fractional nonlinear synergic controller for maximum power point tracking of photovoltaic array under abrupt irradiance change. IEEE J Photovoltaics 2023;13:305–14. https://doi.org/10.1109/JPHOTOV.2023.3236808.

[11] Farag MM, Hamid A-K, Salameh T, Abo-Zahhad EM, Nooman AlMallahi M, Elgendi M. Environmental, economic, and degradation assessment for a 2.88 kW grid-connected PV system under Sharjah weather conditions. In: 50th Int. Conf. Comput. Ind. Eng., 2023, p. 1722–1731.

[12] Shirinbakhsh M, Harvey LDD. Feasibility of achieving net-zero energy performance in high-rise buildings using solar energy. Energy Built Environ 2024;5:946–56. https://doi.org/10.1016/j.enbenv.2023.07.007.

[13] Farag MM, Bansal RC. Solar energy development in the GCC region – a review on recent progress and opportunities. Int J Model Simul 2023;43:579–99. https://doi.org/10.1080/02286203.2022.2105785.

[14] Salameh T, Tawalbeh M, Juaidi A, Abdallah R, Hamid A-K. A novel three-dimensional numerical model for PV/T water system in hot climate region. Renew Energy 2021;164:1320–33. https://doi.org/10.1016/j.renene.2020.10.137.

[15] Salameh T, Zhang D, Juaidi A, Alami AH, Al-Hinti I, Olabi AG. Review of solar photovoltaic cooling systems technologies with environmental and economical assessment. J Clean Prod 2021;326:129421. https://doi.org/10.1016/j.jclepro.2021.129421.

[16] Salamah T, Ramahi A, Alamara K, Juaidi A, Abdallah R, Abdelkareem MA, et al. Effect of dust and methods of cleaning on the performance of solar PV module for different climate regions: Comprehensive review. Sci Total Environ 2022;827:154050. https://doi.org/10.1016/j.scitotenv.2022.154050.

[17] Vikas, Pandey A, Yadav A. Experimental investigation of photovoltaic panel cooling by uniformly flowing water on the top surface for efficiency improvement, 2024, p. 363–76. https://doi.org/10.1007/978-981-99-7213-5_28.

[18] Esen M, Ayhan T. Development of a model compatible with solar assisted cylindrical energy storage tank and variation of stored energy with time for different phase change materials. Energy Convers Manag 1996;37:1775–85. https://doi.org/10.1016/0196-8904(96)00035-0.

[19] Esen M, Durmuş A, Durmuş A. Geometric design of solar-aided latent heat store depending on various parameters and phase change materials. Sol Energy 1998;62:19–28. https://doi.org/10.1016/S0038-092X(97)00104-7.

[20] Oflaz F, Keklikcioglu O, Ozceyhan V. Investigating thermal performance of combined use of SiO2-water nanofluid and newly designed conical wire inserts. Case Stud Therm Eng 2022;38:102378. https://doi.org/10.1016/j.csite.2022.102378.

[21] Alzaabi AA, Badawiyeh NK, Hantoush HO, Hamid AK. Electrical/thermal performance of hybrid PV/T system in Sharjah, UAE. Int J Smart Grid Clean Energy 2014. https://doi.org/10.12720/sgce.3.4.385-389.

[22] Almarzooqi NK, Ahmad FF, Hamid AK, Ghenai C, Farag MM, Salameh T. Experimental investigation of the effect of optical filters on the performance of the solar photovoltaic system. Energy Rep 2023;9:336–44. https://doi.org/10.1016/j.egyr.2022.12.119.

[23] Rabelo M, Yousuf H, Kim J, Dao V-A, Pham DP, Yi J. Progressive cooling techniques for photovoltaic module efficiency and reliability: Comparative evaluation and optimization. Energy Rep 2022;8:8534–45. https://doi.org/10.1016/j.egyr.2022.06.067.

[24] Shalaby SM, Elfakharany MK, Mujtaba IM, Moharram BM, Abosheiasha HF. Development of an efficient nano-fluid cooling/preheating system for PV-RO water desalination pilot plant. Energy Convers Manag 2022;268:115960. https://doi.org/10.1016/j.enconman.2022.115960.

[25] Mi P, Zhang J, Han Y, Li W. Performance study on photovoltaic thermal heat pump system in typical climate zones. Appl Therm Eng 2024;243:122530. https://doi.org/10.1016/j.applthermaleng.2024.122530.

[26] Esen H, Inalli M, Sengur A, Esen M. Modelling a ground-coupled heat pump system using adaptive neuro-fuzzy inference systems. Int J Refrig 2008;31:65–74. https://doi.org/10.1016/j.ijrefrig.2007.06.007.

[27] Esen H, Inalli M, Sengur A, Esen M. Predicting performance of a ground-source heat pump system using fuzzy weighted pre-processing-based ANFIS. Build Environ 2008;43:2178–87. https://doi.org/10.1016/j.buildenv.2008.01.002.

[28] Esen M. Thermal performance of a solar-aided latent heat store used for space heating by heat pump. Sol Energy 2000;69:15–25. https://doi.org/10.1016/S0038-092X(00)00015-3.

[29] Esen H, Ozgen F, Esen M, Sengur A. Modelling of a new solar air heater through least-squares support vector machines. Expert Syst Appl 2009;36:10673–82. https://doi.org/10.1016/j.eswa.2009.02.045.

[30] Esen H, Ozgen F, Esen M, Sengur A. Artificial neural network and wavelet neural network approaches for modelling of a solar air heater. Expert Syst Appl 2009;36:11240–8. https://doi.org/10.1016/j.eswa.2009.02.073.

[31] Esen H, Esen M, Ozsolak O. Modelling and experimental performance analysis of solar-assisted ground source heat pump system. J Exp Theor Artif Intell 2017;29:1–17. https://doi.org/10.1080/0952813X.2015.1056242.

[32] Esen H, Inalli M, Sengur A, Esen M. Modeling a ground-coupled heat pump system by a support vector machine. Renew Energy 2008;33:1814–23. https://doi.org/10.1016/j.renene.2007.09.025.

[33] Salameh T, Hamid AK, Farag MM, Abo-Zahhad EM. Energy and exergy assessment for a University of Sharjah's PV grid-connected system based on experimental for harsh terrestrial conditions. Energy Rep 2023;9:345–53. https://doi.org/10.1016/j.egyr.2022.12.117.

[34] Alktranee M, Shehab MA, Németh Z, Bencs P, Hernadi K, Koós T. Energy and exergy assessment of photovoltaic-thermal system using tungsten trioxide nanofluid: An experimental study. Int J Thermofluids 2022;16:100228. https://doi.org/10.1016/j.ijft.2022.100228.

[35] Salameh T, Hamid AK, Farag MM, Abo-Zahhad EM. Experimental and numerical simulation of a 2.88 kW PV grid-connected system under the terrestrial conditions of Sharjah city. Energy Rep 2023;9:320–7. https://doi.org/10.1016/j.egyr.2022.12.115.

[36] Alktranee M, Bencs P. Experimental comparative study on using different cooling techniques with photovoltaic modules. J Therm Anal Calorim 2023;148:3805–17. https://doi.org/10.1007/s10973-022-11940-1.

[37] Diwania S, Agrawal S, Siddiqui AS, Singh S. Photovoltaic–thermal (PV/T) technology: a comprehensive review on applications and its advancement. Int J Energy Environ Eng 2020;11:33–54. https://doi.org/10.1007/s40095-019-00327-y.

[38] Sohani A, Cornaro C, Shahverdian MH, Hoseinzadeh S, Moser D, Nastasi B, et al. Thermography and machine learning combination for comprehensive analysis of transient response of a photovoltaic module to water cooling. Renew Energy 2023;210:451–61. https://doi.org/10.1016/j.renene.2023.04.073.

[39] Farag MM, Ahmad FF, Hamid AK, Ghenai C, Bettayeb M. Real-time monitoring and performance harvesting for grid-connected PV system - A case in Sharjah. In: 2021 14th Int. Conf. Dev. eSystems Eng., IEEE; 2021, p. 241–5. https://doi.org/10.1109/DeSE54285.2021.9719385.

[40] Garcia Noxpanco M, Wilkins J, Riffat S. A review of the recent development of photovoltaic/thermal (PV/T) systems and their applications. Futur Cities Environ 2020:6. https://doi.org/10.5334/fce.97.

[41] Chavarría-Domínguez B, De León-Aldaco SE, Velázquez-Limón N, Ponce-Silva M, Aguilar-Jiménez JA, Chavarría-Domínguez F. A review of the modeling of parabolic trough solar collectors coupled to solar receivers with photovoltaic/thermal generation. Energies 2024;17:1582. https://doi.org/10.3390/en17071582.

[42] Mateo Romero HF, González Rebollo MÁ, Cardeñoso-Payo V, Alonso Gómez V, Redondo Plaza A, Moyo RT, et al. Applications of artificial intelligence to photovoltaic systems: A review. Appl Sci 2022;12:10056. https://doi.org/10.3390/app121910056.

[43] Chen Q, Li X, Zhang Z, Zhou C, Guo Z, Liu Z, et al. Remote sensing of photovoltaic scenarios: Techniques, applications and future directions. Appl Energy 2023;333:120579. https://doi.org/10.1016/j.apenergy.2022.120579.

[44] Farag MM, Hamid AK. Experimental investigation on the annual performance of an actively monitored 2.88 kW grid-connected PV system in Sharjah, UAE. In: 2023 Adv. Sci. Eng. Technol. Int. Conf., IEEE; 2023, p. 1–6. https://doi.org/10.1109/ASET56582.2023.10180880.

[45] Po-Ching Hwang H, Ku C-C-Y, Chao-Yang HM. Intelligent cleanup scheme for soiled photovoltaic modules. Energy 2023;265:126293. https://doi.org/10.1016/j.energy.2022.126293.

[46] Mellit A, Kalogirou S. Artificial intelligence and internet of things to improve efficacy of diagnosis and remote sensing of solar photovoltaic systems: Challenges, recommendations and future directions. Renew Sustain Energy Rev 2021;143:110889. https://doi.org/10.1016/j.rser.2021.110889.

[47] Shaik F, Lingala SS, Veeraboina P. Effect of various parameters on the performance of solar PV power plant: a review and the experimental study. Sustain Energy Res 2023;10:6. https://doi.org/10.1186/s40807-023-00076-x.

[48] Evans DL, Florschuetz LW. Terrestrial concentrating photovoltaic power system studies. Sol Energy 1978;20:37–43. https://doi.org/10.1016/0038-092X(78)90139-1.

[49] Jakhar S, Soni MS, Gakkhar N. Performance analysis of photovoltaic panels with earth water heat exchanger cooling. MATEC Web Conf 2016;55:02003. https://doi.org/10.1051/matecconf/20165502003.

[50] Kim J, Rabelo M, Padi SP, Yousuf H, Cho E-C, Yi J. A review of the degradation of photovoltaic modules for life expectancy. Energies 2021;14:4278. https://doi.org/10.3390/en14144278.
[51] Farag MM, Faraz Ahmad F, Hamid AK, Ghenai C, AlMallahi M, Elgendi M. Impact of colored filters on PV modules performance: an experimental investigation on electrical and spectral characteristics. In: 50th Int. Conf. Comput. Ind. Eng., 2023, p. 1692–704.
[52] Perin Gasparin F, Detzel Kipper F, Schuck de Oliveira F, Krenzinger A. Assessment on the variation of temperature coefficients of photovoltaic modules with solar irradiance. Sol Energy 2022;244:126–33. https://doi.org/10.1016/j.solener.2022.08.052.
[53] Ishii T, Otani K, Takashima T, Kawai S. Estimation of the maximum power temperature coefficients of PV modules at different time scales. Sol Energy Mater Sol Cells 2011;95:386–9. https://doi.org/10.1016/j.solmat.2010.04.041.
[54] Ahmad FF, Ghenai C, Hamid AK, Rejeb O, Bettayeb M. Performance enhancement and infra-red (IR) thermography of solar photovoltaic panel using back cooling from the waste air of building centralized air conditioning system. Case Stud Therm Eng 2021;24:100840. https://doi.org/10.1016/j.csite.2021.100840.
[55] Dubey S, Sarvaiya JN, Seshadri B. Temperature dependent photovoltaic (PV) efficiency and its effect on PV production in the world – A review. Energy Procedia 2013;33:311–21. https://doi.org/10.1016/j.egypro.2013.05.072.
[56] Hossain R, Ahmed AJ, Islam SMKN, Saha N, Debnath P, Kouzani AZ, et al. New design of solar photovoltaic and thermal hybrid system for performance improvement of solar photovoltaic. Int J Photoenergy 2020;2020:1–6. https://doi.org/10.1155/2020/8825489.
[57] Evans DL. Simplified method for predicting photovoltaic array output. Sol Energy 1981;27:555–60. https://doi.org/10.1016/0038-092X(81)90051-7.
[58] Al-Shamkhee D, Al-Aasam AB, Al-Waeli AHA, Abusaibaa GY, Moria H. Passive cooling techniques for ventilation: an updated review. Renew Energy Environ Sustain 2022;7:23. https://doi.org/10.1051/rees/2022011.
[59] Tiwari MK, Mishra V, Dev R, Singh N. Effects of active cooling techniques to improve the overall efficiency of photovoltaic module-An updated review. E3S Web Conf 2023;387:01012. https://doi.org/10.1051/e3sconf/202338701012.
[60] Jin Z, Xu H, Li G, Zhao X, Liu Z, Wu D, et al. Performance study of organic photovoltaic/thermal system with synergistic effect of photocatalytic and thermal catalytic technology. Sol Energy 2024;271:112456. https://doi.org/10.1016/j.solener.2024.112456.
[61] Farag MM, Ahmad FF, Hamid AK, Ghenai C, Bettayeb M, Alchadirchy M. Performance assessment of a hybrid PV/T system during winter season under Sharjah climate. In: 2021 Int. Conf. Electr. Comput. Commun. Mechatronics Eng., IEEE; 2021, p. 1–5. https://doi.org/10.1109/ICECCME52200.2021.9590896.
[62] Alktranee M, Bencs P. Effect of evaporative cooling on photovoltaic module performance. Process Integr Optim Sustain 2022;6:921–30. https://doi.org/10.1007/s41660-022-00268-w.
[63] Farag MM, Hamid AK. Performance assessment of rooftop PV/T systems based on adaptive and smart cooling facility scheme - a case in hot climatic conditions of Sharjah, UAE. In: 3rd Int. Conf. Distrib. Sens. Intell. Syst. (ICDSIS 2022), Institution of Engineering and Technology; 2022, p. 198–207. https://doi.org/10.1049/icp.2022.2448.
[64] Dwivedi P, Sudhakar K, Soni A, Solomin E, Kirpichnikova I. Advanced cooling techniques of P.V. modules: A state of art. Case Stud Therm Eng 2020;21:100674. https://doi.org/10.1016/j.csite.2020.100674.
[65] Rejeb O, Lamrani B, Lamba R, Kousksou T, Salameh T, Jemni A, et al. Numerical investigations of concentrated photovoltaic thermal system integrated with thermoelectric power generator and phase change material. J Energy Storage 2023;62:106820. https://doi.org/10.1016/j.est.2023.106820.
[66] Rejeb O, Shittu S, Li G, Ghenai C, Zhao X, Ménézo C, et al. Comparative investigation of concentrated photovoltaic thermal-thermoelectric with nanofluid cooling. Energy Convers Manag 2021;235:113968. https://doi.org/10.1016/j.enconman.2021.113968.
[67] Hachicha AA, Abo-Zahhad EM, Said Z, Rahman SMA. Numerical and experimental investigations of the electrical and thermal performances of a novel PV thermal system. Renew Energy 2022;195:990–1000. https://doi.org/10.1016/j.renene.2022.06.083.
[68] Said Z, Ahmad FF, Radwan AM, Hachicha AA. New thermal management technique for PV module using Mist/PCM/Husk: An experimental study. J Clean Prod 2023;401:136798. https://doi.org/10.1016/j.jclepro.2023.136798.
[69] Raina G, Sinha S, Saini G, Sharma S, Malik P, Thakur NS. Assessment of photovoltaic power generation using fin augmented passive cooling technique for different climates. Sustain Energy Technol Assessments 2022;52:102095. https://doi.org/10.1016/j.seta.2022.102095.
[70] Chaichan MT, Mahdi MT, Kazem HA, Al-Waeli AHA, Fayad MA, Al-Amiery AA, et al. Modified nano-Fe2O3-paraffin wax for efficient photovoltaic/thermal system in severe weather conditions. Sustainability 2022;14:12015. https://doi.org/10.3390/su141912015.
[71] Kazem HA, Chaichan MT, Al-Waeli AHA, Jarimi H, Ibrahim A, Sopian K. Effect of temperature on the electrical and thermal behaviour of a photovoltaic/thermal system cooled using SiC nanofluid: an experimental and comparison study. Sustainability 2022;14:11897. https://doi.org/10.3390/su141911897.
[72] Abdel-raheim Amr A, Hassan AAM, Abdel-Salam M, El-Sayed AHM. An experiment-based comparison of different cooling methods for photovoltaic modules. Int J Emerg Electr Power Syst 2024. https://doi.org/10.1515/ijeeps-2023-0477.
[73] Sutanto B, Iacovides H, Nasser A, Cioncolini A, Afgan I, Indartono YS, et al. Design and analysis of passively cooled floating photovoltaic systems. Appl Therm Eng 2024;236:121801. https://doi.org/10.1016/j.applthermaleng.2023.121801.
[74] Liu J, Zhou Y, Zhou Z, Du Y, Wang C, Yang X, et al. Passive photovoltaic cooling: advances toward low-temperature operation. Adv Energy Mater 2024;14. https://doi.org/10.1002/aenm.202302662.
[75] Abdel-Raheim Amr A, Hassan AAM, Abdel-Salam M, El-Sayed AHM. Active cooling of a photovoltaic module in hot-ambient temperatures: theory versus experiment. Int J Emerg Electr Power Syst 2024. https://doi.org/10.1515/ijeeps-2023-0398.
[76] Mostakim K, Akbar MR, Islam MA, Islam MK. Integrated photovoltaic-thermal system utilizing front surface water cooling technique: An experimental performance response. Heliyon 2024;10:e25300. https://doi.org/10.1016/j.heliyon.2024.e25300.
[77] Khan A, Shahzad N, Waqas A, Mahmood M, Ali M, Umar S. Unlocking the potential of passive cooling: A comprehensive experimental study of PV/PCM/TEC hybrid system for enhanced photovoltaic performance. J Energy Storage 2024;80:110277. https://doi.org/10.1016/j.est.2023.110277.
[78] Farzan H, Iranmanesh A, Mohsenifard A. Investigation on different cooling strategies for photovoltaic thermal Systems: An experimental study. Appl Therm Eng 2024;250:123537. https://doi.org/10.1016/j.applthermaleng.2024.123537.
[79] Hamid AK, Mbungu NT, Elnady A, Bansal RC, Ismail AA, AlShabi MA. A systematic review of grid-connected photovoltaic and photovoltaic/thermal systems: Benefits, challenges and mitigation. Energy Environ 2022. https://doi.org/10.1177/0958305X221117617.
[80] Allouhi A. A novel grid-connected solar PV-thermal/wind integrated system for simultaneous electricity and heat generation in single family buildings. J Clean Prod 2021;320:128518. https://doi.org/10.1016/j.jclepro.2021.128518.
[81] Bashir SB, Ismail AAA, Elnady A, Farag MM, Hamid A-K, Bansal RC, et al. Modular multilevel converter-based microgrid: A critical review. IEEE Access 2023;11:65569–89. https://doi.org/10.1109/ACCESS.2023.3289829.
[82] Farag MM, Salameh T, Hamid AK, Hussein M. Experimental investigation on the thermal and exergy efficiency for a 2.88 kW Grid connected photovoltaic/thermal system. In: 7th Int. Conf. Renew. ENERGY Gener. Appl., 2024.
[83] Mosalam H. Evaluation study design and operation of a building integrated photovoltaic system. In: 2018 Int. Conf. Smart Grid, IEEE; 2018, p. 195–201. https://doi.org/10.1109/ISGWCP.2018.8634453.
[84] Al-Waeli AHA, Chaichan MT, Kazem HA, Sopian K. Comparative study to use nano-(Al 2 O 3, CuO, and SiC) with water to enhance photovoltaic thermal PV/T collectors. Energy Convers Manag 2017;148:963–73. https://doi.org/10.1016/j.enconman.2017.06.072.
[85] Fudholi A, Farhana Mohd Razali N, Ridwan A, Yendra R, Hartono H, Pani Desvina A, et al. Overview of photovoltaic thermal (PVT) water collector. Int J Power Electron Drive Syst 2018;9:1891. https://doi.org/10.11591/ijpeds.v9.i4.pp1891-1898.
[86] Pang W, Cui Y, Zhang Y, Zhang X, Yan H. Comparisons of photovoltaic modules for their performances based on different substrates. Appl Therm Eng 2019;146:505–14. https://doi.org/10.1016/j.applthermaleng.2018.10.033.
[87] Maghrabie HM, Mohamed ASA, Fahmy AM, Abdel Samee AA. Performance enhancement of PV panels using phase change material (PCM): An experimental implementation. Case Stud Therm Eng 2023;42:102741. https://doi.org/10.1016/j.csite.2023.102741.
[88] Wang G, Yang Y, Yu W, Wang T, Zhu T. Performance of an air-cooled photovoltaic/thermal system using micro heat pipe array. Appl Therm Eng 2022;217:119184. https://doi.org/10.1016/j.applthermaleng.2022.119184.
[89] Gomaa MR, Ahmed M, Rezk H. Temperature distribution modeling of PV and cooling water PV/T collectors through thin and thick cooling cross-fined channel box. Energy Rep 2022;8:1144–53. https://doi.org/10.1016/j.egyr.2021.11.061.
[90] Esen M. Thermal performance of a solar cooker integrated vacuum-tube collector with heat pipes containing different refrigerants. Sol Energy 2004;76:751–7. https://doi.org/10.1016/j.solener.2003.12.009.
[91] Khelifa A, Touafek K, Ben Moussa H, Tabet I. Modeling and detailed study of hybrid photovoltaic thermal (PV/T) solar collector. Sol Energy 2016;135:169–76. https://doi.org/10.1016/j.solener.2016.05.048.
[92] Gholami A, Ameri M, Zandi M, Ghoachani RG, Pierfederici S, Kazem HA. Step-by-step guide to model photovoltaic panels: an up-to-date comparative review study. IEEE J Photovoltaics 2022;12:915–28. https://doi.org/10.1109/JPHOTOV.2022.3169525.
[93] Yan B, Wu Q, Chi X, Wu C, Luo P, Luo Y, et al. Numerical and experimental investigation of photovoltaic/thermal systems: parameter analysis and determination of optimum flow. Sustainability 2022;14:10156. https://doi.org/10.3390/su141610156.
[94] Al-Waeli AHA, Kazem HA, Yousif JH, Chaichan MT, Sopian K. Mathematical and neural network modeling for predicting and analyzing of nanofluid-nano PCM photovoltaic thermal systems performance. Renew Energy 2020;145:963–80. https://doi.org/10.1016/j.renene.2019.06.099.
[95] Christ D, Kaltschmitt M. Modelling of photovoltaic-thermal collectors for the provision of electricity and low temperature heat—Comparison of different flow rate control approaches to optimize the electrical yield. Renew Energy Focus 2021;37:1–13. https://doi.org/10.1016/j.ref.2021.02.002.
[96] Assareh E, Jafarian M, Nedaei M, Firoozzadeh M, Lee M. Performance evaluation and optimization of a photovoltaic/thermal (PV/T) system according to climatic conditions. Energies 2022;15:7489. https://doi.org/10.3390/en15207489.
[97] Yaghoubi AA, Gandomzadeh M, Gholami A, Gavagsaz Ghoachani R, Zandi M, Kazem AH. Optimize photovoltaic panels cleaning scheduling framework based on variations of hourly-based active electricity pricing in the market. Sol Energy 2024;275:112633. https://doi.org/10.1016/j.solener.2024.112633.

[98] Abril-Macias G, Peralta-Jaramillo J, Delgado-Plaza E, Sosa-Tinoco I, Avilés D. Simulation of the fluid dynamic and thermal behavior of an experimental passive cooling system of photovoltaic panels. Heliyon 2024;10:e24768. https://doi.org/10.1016/j.heliyon.2024.e24768.

[99] Bugeja R, Mule' Stagno L, Niarchos I. Full-scale design, implementation and testing of an innovative photovoltaic cooling system (IPCoSy). Sustainability 2023;15:16900. https://doi.org/10.3390/su152416900.

[100] Abd El-Hamid M, Wei G, Cui L, Xu C, Du X. Three-dimensional heat transfer studies of glazed and unglazed Photovoltaic/Thermal systems embedded with phase change materials. Appl Therm Eng 2022;208:118222. https://doi.org/10.1016/j.applthermaleng.2022.118222.

[101] Parthiban A, Reddy KS, Pesala B, Mallick TK. Effects of operational and environmental parameters on the performance of a solar photovoltaic-thermal collector. Energy Convers Manag 2020;205:112428. https://doi.org/10.1016/j.enconman.2019.112428.

[102] Oztop HF, Sahin AZ, Coşanay H, Sahin IH. Three-dimensional computational analysis of performance improvement in a novel designed solar photovoltaic/thermal system by using hybrid nanofluids. Renew Energy 2023;210:832–41. https://doi.org/10.1016/j.renene.2023.04.115.

[103] Kalkan C, Duquette J, Akif EM. Development of a novel computational fluid dynamics-based model for a solar photovoltaic/thermal collector-assisted domestic hot water system with sensible heat storage. Appl Therm Eng 2023;228:120424. https://doi.org/10.1016/j.applthermaleng.2023.120424.

[104] Kuo C-F-J, Liu J-M, Umar ML, Lan W-L, Huang C-Y, Syu S-S. The photovoltaic-thermal system parameter optimization design and practical verification. Energy Convers Manag 2019;180:358–71. https://doi.org/10.1016/j.enconman.2018.10.080.

[105] Kalkan C, Ezan MA, Duquette J, Yilmaz Balaman Ş, Yilanci A. Numerical study on photovoltaic/thermal systems with extended surfaces. Int J Energy Res 2019;43:5213–29. https://doi.org/10.1002/er.4477.

[106] Arefin MA. Analysis of an integrated photovoltaic thermal system by top surface natural circulation of water. Front Energy Res 2019:7. https://doi.org/10.3389/fenrg.2019.00097.

[107] Khordehgah N, Żabnieńska-Góra A, Jouhara H. Analytical modelling of a photovoltaics-thermal technology combined with thermal and electrical storage systems. Renew Energy 2021;165:350–8. https://doi.org/10.1016/j.renene.2020.11.058.

[108] Al-Amri FG, Mohammed Abdelmagid TI. Analytical model for the prediction of solar cell temperature for a high-concentration photovoltaic system. Case Stud Therm Eng 2021;25:100890. https://doi.org/10.1016/j.csite.2021.100890.

[109] Barbu M, Darie G, Siroux M. Analysis of a residential photovoltaic-thermal (PVT) system in two similar climate conditions. Energies 2019;12:3595. https://doi.org/10.3390/en12193595.

[110] Hocine cheikh el HB, Touafek K, Kerrour F, Haloui H, Khelifa A. Model Validation of an Empirical Photovoltaic Thermal (PV/T) Collector. Energy Procedia 2015;74:1090–9. https://doi.org/10.1016/j.egypro.2015.07.749.

[111] Hassan O, Zakzouk N, Abdelsalam A. Novel photovoltaic empirical mathematical model based on function representation of captured figures from commercial panels datasheet. Mathematics 2022;10:476. https://doi.org/10.3390/math10030476.

[112] Jatoi AR, Samo SR, Jakhrani AQ. An improved empirical model for estimation of temperature effect on performance of photovoltaic modules. Int J Photoenergy 2019;2019:1–16. https://doi.org/10.1155/2019/1681353.

[113] Mojumder JC, Ong HC, Chong WT, Leong KY, Izadyar N. An empirical analysis on photovoltaic thermal system with fin design by forced air circulation. J Mech Sci Technol 2017;31:2549–57. https://doi.org/10.1007/s12206-017-0453-1.

[114] Kahani M, Zamen M, Rostami B. Modeling and empirical study of TiO2/water nanofluid flows in a modified configuration with new layer arrangement of a photovoltaic/thermal system. Sustain Energy Technol Assessments 2022;51:101932. https://doi.org/10.1016/j.seta.2021.101932.

[115] Vaziri Rad MA, Kasaeian A, Mousavi S, Rajaee F, Kouravand A. Empirical investigation of a photovoltaic-thermal system with phase change materials and aluminum shavings porous media. Renew Energy 2021;167:662–75. https://doi.org/10.1016/j.renene.2020.11.135.

[116] Qingyang J, Jichun Y, Yanying Z, Huide F. Energy and exergy analyses of PV, solar thermal and photovoltaic/thermal systems: a comparison study. Int J Low-Carbon Technol 2021;16:604–11. https://doi.org/10.1093/ijlct/ctaa092.

[117] Gholami A, Ameri M, Zandi M, Gavagsaz Ghoachani R, Kazem HA. Predicting solar photovoltaic electrical output under variable environmental conditions: Modified semi-empirical correlations for dust. Energy Sustain Dev 2022;71:389–405. https://doi.org/10.1016/j.esd.2022.10.012.

[118] Zhang T, Zheng W, Wang L, Yan Z, Hu M. Experimental study and numerical validation on the effect of inclination angle to the thermal performance of solar heat pipe photovoltaic/thermal system. Energy 2021;223:120020. https://doi.org/10.1016/j.energy.2021.120020.

[119] Gholami A, Ameri M, Zandi M, Gavagsaz GR. A single-diode model for photovoltaic panels in variable environmental conditions: Investigating dust impacts with experimental evaluation. Sustain Energy Technol Assessments 2021;47:101392. https://doi.org/10.1016/j.seta.2021.101392.

[120] Gholami A, Ameri M, Zandi M, Gavagsaz Ghoachani R, Jafarzadegan Gerashi S, Kazem HA, et al. Impact of harsh weather conditions on solar photovoltaic cell temperature: Experimental analysis and thermal-optical modeling. Sol Energy 2023;252:176–94. https://doi.org/10.1016/j.solener.2023.01.039.

[121] Chaibi Y, El Rhafiki T, Simón-Allué R, Guedea I, Luaces C, Charro Gajate O, et al. Physical models for the design of photovoltaic/thermal collector systems. Sol Energy 2021;226:134–46. https://doi.org/10.1016/j.solener.2021.08.048.

[122] Sredenšek K, Seme S, Štumberger B, Hadžiselimović M, Chowdhury A, Praunseis Z. Experimental validation of a dynamic photovoltaic/thermal collector model in combination with a thermal energy storage tank. Energies 2021;14:8162. https://doi.org/10.3390/en14238162.

[123] Sarkar MNI. Effect of various model parameters on solar photovoltaic cell simulation: a SPICE analysis. Renewables Wind Water, Sol 2016;3:13. https://doi.org/10.1186/s40807-016-0035-3.

[124] Shockley W. The theory of p-n junctions in semiconductors and p-n junction transistors. Bell Syst Tech J 1949;28:435–89. https://doi.org/10.1002/j.1538-7305.1949.tb03645.x.

[125] Chin VJ, Salam Z, Ishaque K. Cell modelling and model parameters estimation techniques for photovoltaic simulator application: A review. Appl Energy 2015;154:500–19. https://doi.org/10.1016/j.apenergy.2015.05.035.

[126] Khezzar R, Zereg M, Khezzar A. Modeling improvement of the four parameter model for photovoltaic modules. Sol Energy 2014;110:452–62. https://doi.org/10.1016/j.solener.2014.09.039.

[127] Chaibi Y, Salhi M, El-jouni A, Essadki A. A new method to extract the equivalent circuit parameters of a photovoltaic panel. Sol Energy 2018;163:376–86. https://doi.org/10.1016/j.solener.2018.02.017.

[128] Chaibi Y, Malvoni M, Allouhi A, Mohamed S. Data on the I-V characteristics related to the SM55 monocrystalline PV module at various solar irradiance and temperatures. Data Br 2019;26:104527. https://doi.org/10.1016/j.dib.2019.104527.

[129] Chaibi Y, Allouhi A, Salhi M. A simple iterative method to determine the electrical parameters of photovoltaic cell. J Clean Prod 2020;269:122363. https://doi.org/10.1016/j.jclepro.2020.122363.

[130] Chan DSH, Phang JCH. Analytical methods for the extraction of solar-cell single- and double-diode model parameters from I-V characteristics. IEEE Trans Electron Devices 1987;34:286–93. https://doi.org/10.1109/T-ED.1987.22920.

[131] Lo Brano V, Orioli A, Ciulla G, Di Gangi A. An improved five-parameter model for photovoltaic modules. Sol Energy Mater Sol Cells 2010;94:1358–70. https://doi.org/10.1016/j.solmat.2010.04.003.

[132] Maria Y. Machine learning based approaches for modeling the output power of photovoltaic array in real outdoor conditions. Electronics 2020;9:315. https://doi.org/10.3390/electronics9020315.

[133] Pillai DS, Rajasekar N. Metaheuristic algorithms for PV parameter identification: A comprehensive review with an application to threshold setting for fault detection in PV systems. Renew Sustain Energy Rev 2018;82:3503–25. https://doi.org/10.1016/j.rser.2017.10.107.

[134] Ishaque K, Salam Z, Taheri H. Simple, fast and accurate two-diode model for photovoltaic modules. Sol Energy Mater Sol Cells 2011;95:586–94. https://doi.org/10.1016/j.solmat.2010.09.023.

[135] Kumar S, Sahu HS, Nayak SK. Estimation of MPP of a double diode model PV module from explicit I-V characteristic. IEEE Trans Ind Electron 2019;66:7032–42. https://doi.org/10.1109/TIE.2018.2877116.

[136] Jordehi AR. Parameter estimation of solar photovoltaic (PV) cells: A review. Renew Sustain Energy Rev 2016;61:354–71. https://doi.org/10.1016/j.rser.2016.03.049.

[137] Gao X, Cui Y, Hu J, Xu G, Yu Y. Lambert W-function based exact representation for double diode model of solar cells: Comparison on fitness and parameter extraction. Energy Convers Manag 2016;127:443–60. https://doi.org/10.1016/j.enconman.2016.09.005.

[138] Ishaque K, Salam Z, Taheri H, Syafaruddin. Modeling and simulation of photovoltaic (PV) system during partial shading based on a two-diode model. Simul Model Pract Theory 2011;19:1613–26. https://doi.org/10.1016/j.simpat.2011.04.005.

[139] Pandey PK, Sandhu KS. Multi diode modelling of PV cell. In: 2014 IEEE 6th India Int. Conf. Power Electron., IEEE; 2014, p. 1–4. https://doi.org/10.1109/IICPE.2014.7115793.

[140] Qais MH, Hasanien HM, Alghuwainem S, Nouh AS. Coyote optimization algorithm for parameters extraction of three-diode photovoltaic models of photovoltaic modules. Energy 2019;187:116001. https://doi.org/10.1016/j.energy.2019.116001.

[141] Allam D, Yousri DA, Eteiba MB. Parameters extraction of the three diode model for the multi-crystalline solar cell/module using Moth-Flame Optimization Algorithm. Energy Convers Manag 2016;123:535–48. https://doi.org/10.1016/j.enconman.2016.06.052.

[142] Khanna V, Das BK, Bisht D, Vandana SPK. A three diode model for industrial solar cells and estimation of solar cell parameters using PSO algorithm. Renew Energy 2015;78:105–13. https://doi.org/10.1016/j.renene.2014.12.072.

[143] Bechouat M, Sedraoui M, Feraga C-E, Aidoud M, Kahla S. Modeling and fuzzy MPPT controller design for photovoltaic module equipped with a closed-loop cooling system. J Electron Mater 2019;48:5471–80. https://doi.org/10.1007/s11664-019-07243-1.

[144] Slimani MEA, Amirat M, Kurucz I, Bahria S, Hamidat A, Chaouch WB. A detailed thermal-electrical model of three photovoltaic/thermal (PV/T) hybrid air collectors and photovoltaic (PV) module: Comparative study under Algiers climatic conditions. Energy Convers Manag 2017;133:458–76. https://doi.org/10.1016/j.enconman.2016.10.066.

[145] Catelani M, Ciani L, Kazimierczuk MK, Reatti A. Matlab PV solar concentrator performance prediction based on triple junction solar cell model. Measurement 2016;88:310–7. https://doi.org/10.1016/j.measurement.2016.03.046.

[146] Babu C, Ponnambalam P. The theoretical performance evaluation of hybrid PV-TEG system. Energy Convers Manag 2018;173:450–60. https://doi.org/10.1016/j.enconman.2018.07.104.

[147] Waliullah M, Hossain MZ, Saha S. On the implementation of two-diode model for photovoltaic-thermal systems. Procedia Eng 2015;105:725–32. https://doi.org/10.1016/j.proeng.2015.05.063.
[148] Tina G, Cosentino F, Notton G. Effect of thermal gradient on electrical efficiency of hybrid PV/T. In: 25th Eur Photovolt Sol Energy Conf Exhib World Conf Photovolt Energy Convers 2010:6–10.
[149] Farag MM, Alhamad RA, Nassif AB. Metaheuristic algorithms in optimal power flow analysis: A qualitative systematic review. Int J Artif Intell Tools 2023:32. https://doi.org/10.1142/S021821302350032X.
[150] Touati F, Khandakar A, E.H. Chowdhury M, Jr. S.P. Gonzales A, Kim Sorino C, Benhmed K. Photo-Voltaic (PV) Monitoring System, Performance Analysis and Power Prediction Models in Doha, Qatar. Renew. Energy - Technol. Appl., IntechOpen; 2021. https://doi.org/10.5772/intechopen.92632.
[151] Du B, D. Lund P. Application of Artificial Neural Network in Solar Energy, 2023. https://doi.org/10.5772/intechopen.106977.
[152] Omar AA, Farag MM, Alhamad RA. Artifical intelligence: new paradigm in deep space exploration. In: 2021 14th Int. Conf. Dev. eSystems Eng., IEEE; 2021, p. 438–42. https://doi.org/10.1109/DeSE54285.2021.9719425.
[153] Azizi A, Pleimling M. A cautionary tale for machine learning generated configurations in presence of a conserved quantity. Sci Rep 2021;11:6395. https://doi.org/10.1038/s41598-021-85683-8.
[154] Çolak AB. A novel comparative analysis between the experimental and numeric methods on viscosity of zirconium oxide nanofluid: Developing optimal artificial neural network and new mathematical model. Powder Technol 2021;381:338–51. https://doi.org/10.1016/j.powtec.2020.12.053.
[155] Du B, Lund PD, Wang J. Improving the accuracy of predicting the performance of solar collectors through clustering analysis with artificial neural network models. Energy Rep 2022;8:3970–81. https://doi.org/10.1016/j.egyr.2022.03.013.
[156] Roy K, Mandal KK, Mandal AC. Ant-Lion Optimizer algorithm and recurrent neural network for energy management of micro grid connected system. Energy 2019;167:402–16. https://doi.org/10.1016/j.energy.2018.10.153.
[157] Olabi AG, Abdelkareem MA, Semeraro C, Al RM, Rezk H, Muhaisen O, et al. Artificial neural networks applications in partially shaded PV systems. Therm Sci Eng Prog 2023;37:101612. https://doi.org/10.1016/j.tsep.2022.101612.
[158] Gumar AK, Demir F. Solar photovoltaic power estimation using meta-optimized neural networks. Energies 2022;15:8669. https://doi.org/10.3390/en15228669.
[159] Zheng L, Su R, Sun X, Guo S. Historical PV-output characteristic extraction based weather-type classification strategy and its forecasting method for the day-ahead prediction of PV output. Energy 2023;271:127009. https://doi.org/10.1016/j.energy.2023.127009.
[160] Bashir SB, Farag MM, Hamid AK, Adam AA, Abo-Khalil AG, Bansal R. A novel hybrid CNN-XGBoost model for photovoltaic system power forecasting. In: 2024 6th Int. Youth Conf. Radio Electron. Electr. Power Eng., IEEE; 2024, p. 1–6. https://doi.org/10.1109/REEPE60449.2024.10479878.
[161] Elsheikh AH, Sharshir SW, Abd Elaziz M, Kabeel AE, Guilan W, Haiou Z. Modeling of solar energy systems using artificial neural network: A comprehensive review. Sol Energy 2019;180:622–39. https://doi.org/10.1016/j.solener.2019.01.037.
[162] Bellos E, Tzivanidis C. Development of an analytical model for the daily performance of solar thermal systems with experimental validation. Sustain Energy Technol Assessments 2018;28:22–9. https://doi.org/10.1016/j.seta.2018.05.003.
[163] Salameh T, Farag MM, Hamid AK, Hussein M. Adaptive neuro-fuzzy inference system for DC power forecasting for grid-connected PV system in Sharjah. In: 7th Int. Conf. Renew. ENERGY Gener. Appl., 2024.
[164] Ghritlahre HK, Prasad RK. Application of ANN technique to predict the performance of solar collector systems - A review. Renew Sustain Energy Rev 2018;84:75–88. https://doi.org/10.1016/j.rser.2018.01.001.
[165] Osma-Pinto G, Ordóñez-Plata G. Dynamic thermal modelling for the prediction of the operating temperature of a PV panel with an integrated cooling system. Renew Energy 2020;152:1041–54. https://doi.org/10.1016/j.renene.2020.01.132.
[166] Wang S, Bagherzadeh SA, Abdalla AN, Nazir MS. Prediction of the exergy performance of a hybrid photovoltaic/thermal-thermal wheel system using an optimal artificial neural network. Sustain Energy Technol Assessments 2022;53:102711. https://doi.org/10.1016/j.seta.2022.102711.
[167] Zhang X, Hao Y, Shangguan H, Zhang P, Wang A. Detection of surface defects on solar cells by fusing Multi-channel convolution neural networks. Infrared Phys Technol 2020;108:103334. https://doi.org/10.1016/j.infrared.2020.103334.
[168] Heo J, Jung J, Kim B, Han S. Digital elevation model-based convolutional neural network modeling for searching of high solar energy regions. Appl Energy 2020;262:114588. https://doi.org/10.1016/j.apenergy.2020.114588.
[169] Zhang Y, Liu T, Chen C, Zhou X. Visual form perception supports approximate number system acuity and arithmetic fluency. Learn Individ Differ 2019;71:1–12. https://doi.org/10.1016/j.lindif.2019.02.008.
[170] Wang K, Qi X, Liu H. A comparison of day-ahead photovoltaic power forecasting models based on deep learning neural network. Appl Energy 2019;251:113315. https://doi.org/10.1016/j.apenergy.2019.113315.
[171] Zhang Q, Zhang M, Chen T, Sun Z, Ma Y, Yu B. Recent advances in convolutional neural network acceleration. Neurocomputing 2019;323:37–51. https://doi.org/10.1016/j.neucom.2018.09.038.
[172] Gu J, Wang Z, Kuen J, Ma L, Shahroudy A, Shuai B, et al. Recent advances in convolutional neural networks. Pattern Recognit 2018;77:354–77. https://doi.org/10.1016/j.patcog.2017.10.013.
[173] Serrano-Luján L, Toledo C, Colmenar JM, Abad J, Urbina A. Accurate thermal prediction model for building-integrated photovoltaics systems using guided artificial intelligence algorithms. Appl Energy 2022;315:119015. https://doi.org/10.1016/j.apenergy.2022.119015.
[174] Alnaqi AA, Moayedi H, Shahsavar A, Nguyen TK. Prediction of energetic performance of a building integrated photovoltaic/thermal system thorough artificial neural network and hybrid particle swarm optimization models. Energy Convers Manag 2019;183:137–48. https://doi.org/10.1016/j.enconman.2019.01.005.
[175] Diwania S, Kumar M, Kumar R, Kumar A, Gupta V, Khetrapal P. Machine learning-based thermo-electrical performance improvement of nanofluid-cooled photovoltaic–thermal system. Energy Environ 2022:. https://doi.org/10.1177/0958305X221146947.
[176] Cao Y, Kamrani E, Mirzaei S, Khandakar A, Vaferi B. Electrical efficiency of the photovoltaic/thermal collectors cooled by nanofluids: Machine learning simulation and optimization by evolutionary algorithm. Energy Rep 2022;8:24–36. https://doi.org/10.1016/j.egyr.2021.11.252.
[177] Karimi M, Aminzadehsarikhanbeglou E, Vaferi B. Robust intelligent topology for estimation of heat capacity of biochar pyrolysis residues. Measurement 2021;183:109857. https://doi.org/10.1016/j.measurement.2021.109857.
[178] Esmaeili-Faraj SH, Vaferi B, Bolhasani A, Karamian S, Hosseini S, Rashedi R. Design of a neuro-based computing paradigm for simulation of industrial olefin plants. Chem Eng Technol 2021;44:1382–9. https://doi.org/10.1002/ceat.202000442.
[179] Rani DKS, Anila DS. Comparison of feed forward, Cascade forward and Layer Recurrent Algorithm model for breast cancer prediction. IOP Conf Ser Mater Sci Eng 2019;705:012055. https://doi.org/10.1088/1757-899X/705/1/012055.
[180] Satapathy SK, Dehuri S, Jagadev AK, Mishra S. EEG signal classification using RBF neural network trained with improved PSO algorithm for epilepsy identification. In: EEG Brain Signal Classif. Epileptic Seizure Disord. Detect. Elsevier; 2019. p. 67–89. https://doi.org/10.1016/B978-0-12-817426-5.00004-1.
[181] Zhong J, Liu L, Sun Q, Wang X. Prediction of photovoltaic power generation based on general regression and back propagation neural network. Energy Procedia 2018;152:1224–9. https://doi.org/10.1016/j.egypro.2018.09.173.
[182] Nabavi M, Nazarpour V, Alibak AH, Bagherzadeh A, Alizadeh SM. Smart tracking of the influence of alumina nanoparticles on the thermal coefficient of nanosuspensions: application of LS-SVM methodology. Appl Nanosci 2021;11:2113–28. https://doi.org/10.1007/s13204-021-01949-7.
[183] Varol Y, Koca A, Oztop HF, Avci E. Forecasting of thermal energy storage performance of Phase Change Material in a solar collector using soft computing techniques. Expert Syst Appl 2010;37:2724–32. https://doi.org/10.1016/j.eswa.2009.08.007.
[184] Ahmadi MH, Baghban A, Salwana E, Sadeghzadeh M, Zamen M, Shamshirband6 S, et al. Machine learning prediction Models of Electrical Efficiency of Photovoltaic-Thermal Collectors 2019. https://doi.org/10.20944/preprints201905.0033.v1.
[185] Fu C fang, Ji Y, Alazzawi A k, Lu M, Zhao B, Luo Q. Utilization of least squares support vector machine for predicting the yearly exergy yield of a hybrid renewable energy system composed of a building integrated photovoltaic thermal system and an earth air heat exchanger system. Eng Anal Bound Elem 2023;152:293–300. https://doi.org/10.1016/j.enganabound.2023.03.045.
[186] Khosravi K, Mohammed HI, Mahdi JM, Silakhori M, Talebizadeh SP. Developing a predictive model and multi-objective optimization of a photovoltaic/thermal system based on energy and exergy analysis using response surface methodology. SSRN Electron J 2022. https://doi.org/10.2139/ssrn.4171638.
[187] Alsarraf J, Moayedi H, Rashid ASA, Muazu MA, Shahsavar A. Application of PSO–ANN modelling for predicting the exergetic performance of a building integrated photovoltaic/thermal system. Eng Comput 2020;36:633–46. https://doi.org/10.1007/s00366-019-00721-4.
[188] Gao W, Moayedi H, Shahsavar A. The feasibility of genetic programming and ANFIS in prediction energetic performance of a building integrated photovoltaic thermal (BIPVT) system. Sol Energy 2019;183:293–305. https://doi.org/10.1016/j.solener.2019.03.016.
[189] Kalani H, Sardarabadi M, Passandideh-Fard M. Using artificial neural network models and particle swarm optimization for manner prediction of a photovoltaic thermal nanofluid based collector. Appl Therm Eng 2017;113:1170–7. https://doi.org/10.1016/j.applthermaleng.2016.11.105.
[190] Zamen M, Baghban A, Pourkiaei SM, Ahmadi MH. Optimization methods using artificial intelligence algorithms to estimate thermal efficiency of PV/T system. Energy Sci Eng 2019;7:821–34. https://doi.org/10.1002/ese3.312.
[191] Sohani A, Dehnavi A, Sayyaadi H, Hoseinzadeh S, Goodarzi E, Garcia DA, et al. The real-time dynamic multi-objective optimization of a building integrated photovoltaic thermal (BIPV/T) system enhanced by phase change materials. J Energy Storage 2022;46:103777. https://doi.org/10.1016/j.est.2021.103777.
[192] Shaban WM, Kabeel AE, El Hadi AM, Talaat FM. Optimizing photovoltaic thermal solar systems efficiency through advanced artificial intelligence driven thermal management techniques. Appl Therm Eng 2024;247:123029. https://doi.org/10.1016/j.applthermaleng.2024.123029.
[193] Javadijam R, Dehbashi M, Shahverdian MH, Sohani A, Arıcı M, Sayyaadi H. Artificial intelligent based techno-economic-exergetic optimization of a thermoelectric enhanced building integrated photovoltaic thermal system. J Build Eng 2024;84:108526. https://doi.org/10.1016/j.jobe.2024.108526.
[194] Shboul B, Zayed ME, Ashraf WM, Usman M, Roy D, Irshad K, et al. Energy and economic analysis of building integrated photovoltaic thermal system: Seasonal dynamic modeling assisted with machine learning-aided method and multi-objective genetic optimization. Alexandria Eng J 2024;94:131–48. https://doi.org/10.1016/j.aej.2024.03.049.

[195] Hamada A, Emam M, Refaey HA, Moawed M, Abdelrahman MA, Elsayed MEA. Identification of a different design of a photovoltaic thermal collector based on fuzzy logic control and the ARMAX model. Therm Sci Eng Prog 2024;48:102395. https://doi.org/10.1016/j.tsep.2024.102395.
[196] Deka MJ, Kamble AD, Das D, Sharma P, Ali S, Kalita P, et al. Enhancing the performance of a photovoltaic thermal system with phase change materials: Predictive modelling and evaluation using neural networks. Renew Energy 2024; 224:120091. https://doi.org/10.1016/j.renene.2024.120091.
[197] Zhou S, Lu W, Li W, Wang S. Forecasting the temperature of a building-integrated photovoltaic panel equipped with phase change material using artificial neural network. Case Stud Therm Eng 2024;57:104355. https://doi.org/10.1016/j.csite.2024.104355.
[198] Margoum S, Hajji B, Aneli S, Tina GM, Gagliano A. Optimizing nanofluid hybrid solar collectors through artificial intelligence models. Energies 2024;17:2307. https://doi.org/10.3390/en17102307.
[199] Jakhar S, Paliwal MK, Kumar M. Machine learning predictive models for optimal design of photovoltaic/thermal collector with nanofluids based geothermal cooling. Environ Prog Sustain Energy 2023:42. https://doi.org/10.1002/ep.14131.
[200] Yousif JH, Kazem HA. Prediction and evaluation of photovoltaic-thermal energy systems production using artificial neural network and experimental dataset. Case Stud Therm Eng 2021;27:101297. https://doi.org/10.1016/j.csite.2021.101297.
[201] Heidari AA, Faris H, Aljarah I, Mirjalili S. An efficient hybrid multilayer perceptron neural network with grasshopper optimization. Soft Comput 2019;23: 7941–58. https://doi.org/10.1007/s00500-018-3424-2.
[202] Kumar M, Mishra SK. Jaya based functional link multilayer perceptron adaptive filter for Poisson noise suppression from X-ray images. Multimed Tools Appl 2018;77:24405–25. https://doi.org/10.1007/s11042-017-5592-y.
[203] Sharaf M, Yousef MS, Huzayyin AS. Review of cooling techniques used to enhance the efficiency of photovoltaic power systems. Environ Sci Pollut Res 2022;29: 26131–59. https://doi.org/10.1007/s11356-022-18719-9.
[204] Pathak SK, Sharma PO, Goel V, Bhattacharyya S, Aybar HŞ, Meyer JP. A detailed review on the performance of photovoltaic/thermal system using various cooling methods. Sustain Energy Technol Assessments 2022;51:101844. https://doi.org/10.1016/j.seta.2021.101844.
[205] Dixit KK, Yadav I, Gupta GK, Kumar Maurya S. A review on cooling techniques used for photovoltaic panels. In: 2020 Int. Conf. Power Electron. IoT Appl. Renew. Energy its Control, IEEE; 2020, p. 360–4. https://doi.org/10.1109/PARC49193.2020.236626.
[206] Al-Waeli AHA, Sopian K, Kazem HA, Chaichan MT. Design configuration and operational parameters of bi-fluid PVT collectors: an updated review. Environ Sci Pollut Res 2023. https://doi.org/10.1007/s11356-023-25321-0.
[207] Shahsavar A, Alwaeli AHA, Azimi N, Rostami S, Sopian K, Arıcı M, et al. Exergy studies in water-based and nanofluid-based photovoltaic/thermal collectors: Status and prospects. Renew Sustain Energy Rev 2022;168:112740. https://doi.org/10.1016/j.rser.2022.112740.
[208] Ul Abdin Z, Rachid A. A survey on applications of hybrid PV/T panels. Energies 2021;14:1205. https://doi.org/10.3390/en14041205.
[209] Husain SA, Othman M, Khalili NNW. A review on the important key properties of mathematical models describing photovoltaic/thermal (PV/T) solar collectors. System 2022:149–56. https://doi.org/10.1007/978-3-030-79606-8_11.
[210] Sultan SM, Ervina Efzan MN. Review on recent Photovoltaic/Thermal (PV/T) technology advances and applications. Sol Energy 2018;173:939–54. https://doi.org/10.1016/j.solener.2018.08.032.
[211] Chaichan MT, Kazem HA, Al-Waeli AHA, Sopian K, Fayad MA, Alawee WH, et al. Sand and dust storms' impact on the efficiency of the photovoltaic modules installed in Baghdad: A review study with an empirical investigation. Energies 2023;16:3938. https://doi.org/10.3390/en16093938.
[212] Badran A, Obeidat F. Solar hot water heating and electricity generation using PV/T hybrid system. J Ecol Eng 2022;23:196–206. https://doi.org/10.12911/22998993/146783.
[213] Al-Odat M. Experimental study of temperature influence on the performance of PV/T cell under Jordan climate conditions. J Ecol Eng 2022;23:80–8. https://doi.org/10.12911/22998993/152283.
[214] Shalaby SM, Elfakharany MK, Moharram BM, Abosheiasha HF. Experimental study on the performance of PV with water cooling. Energy Rep 2022;8:957–61. https://doi.org/10.1016/j.egyr.2021.11.155.
[215] Paliwal MK, Jakhar S, Sharma V. A comparative study on photovoltaic/thermal systems with various cooling methods. Mater Today Proc 2022. https://doi.org/10.1016/j.matpr.2022.12.115.
[216] Firoozzadeh M, Lotfi M, Shiravi AH, Rajabzadeh DM. An experimental study on using water streaks and water film over PV module to enhance the electrical efficiency. Environ Sci Pollut Res 2023. https://doi.org/10.1007/s11356-023-27117-8.
[217] AlMallahi M, Nooman EH, Assad M, AlShihabi S, Alayi R. Multi-criteria decision-making approach for the selection of cleaning method of solar PV panels in United Arab Emirates based on sustainability perspective. Int J Low-Carbon Technol 2022;17:380–93. https://doi.org/10.1093/ijlct/ctac010.
[218] Kazem HA, Chaichan MT, Al-Waeli AHA, Al-Badi R, Fayad MA, Gholami A. Dust impact on photovoltaic/thermal system in harsh weather conditions. Sol Energy 2022;245:308–21. https://doi.org/10.1016/j.solener.2022.09.012.
[219] Kazem HA, Al-Waeli AHA, Chaichan MT, Sopian K, Gholami A, Alnaser WE. Dust and cleaning impact on the performance of photovoltaic: an outdoor experimental study. Energy Sources, Part A Recover Util Environ Eff 2023;45:3107–24. https://doi.org/10.1080/15567036.2023.2191064.
[220] Hachicha AA, Abo-Zahhad EM. Dust effect on solar energy systems and mitigation methods. Int J Energy Prod Manag 2023;8:97–105. https://doi.org/10.18280/ijepm.080206.
[221] Li D, King M, Dooner M, Guo S, Wang J. Study on the cleaning and cooling of solar photovoltaic panels using compressed airflow. Sol Energy 2021;221:433–44. https://doi.org/10.1016/j.solener.2021.04.050.
[222] Lebbi M, Touafek K, Benchatti A, Boutina L, Khelifa A, Baissi MT, et al. Energy performance improvement of a new hybrid PV/T Bi-fluid system using active cooling and self-cleaning: Experimental study. Appl Therm Eng 2021;182: 116033. https://doi.org/10.1016/j.applthermaleng.2020.116033.
[223] Aljaghoub H, Abumadi F, AlMallahi MN, Obaideen K, Alami AH. Solar PV cleaning techniques contribute to Sustainable Development Goals (SDGs) using Multi-criteria decision-making (MCDM): Assessment and review. Int J Thermofluids 2022;16:100233. https://doi.org/10.1016/j.ijft.2022.100233.
[224] Nualboonrueng T, Tuenpusa P, Ueda Y, Akisawa A. Field experiments of PV-thermal collectors for residential application in Bangkok. Energies 2012;5: 1229–44. https://doi.org/10.3390/en5041229.
[225] Halwani S, Farag MM, Hamid AK, Hussein M. An automated and cost-efficient method for photovoltaic dust cleaning based on biaxially oriented polyamide coating material. In: 7th Int. Conf. Renew. Energy Gener. Appl., 2024.
[226] Nallapaneni S, Shah K, Dhiman HS. Automated solar PV array cleaning based on aerial computer vision. Framework 2023:563–71. https://doi.org/10.1007/978-981-19-9858-4_48.
[227] Li T, Yang D, Chen C, Zeng Z, Huang G, Tao B, et al. A mobile robot design for efficient and large-scale solar panel cleaning. In: 2022 IEEE Int. Conf. Robot. Biomimetics, IEEE; 2022, p. 70–5. https://doi.org/10.1109/ROBIO55434.2022.10011850.
[228] Amin A, Wang X, Alroichdi A, Ibrahim A. Designing and manufacturing a robot for dry-cleaning PV solar panels. Int J Energy Res 2023;2023:1–15. https://doi.org/10.1155/2023/7231554.
[229] Ansari S, Ayob A, Lipu MSH, Saad MHM, Hussain A. A review of monitoring technologies for solar PV systems using data processing modules and transmission protocols: progress, challenges and prospects. Sustainability 2021;13:8120. https://doi.org/10.3390/su13158120.
[230] Mishra SK. Monitoring of PV panels and measurement system for solar. Int J Res Appl Sci Eng Technol 2021;9:2447–53. https://doi.org/10.22214/ijraset.2021.34820.
[231] Halwani S, Farag MM, Hamid A-K, Ahmad FF, Ghenai C. A robust method for diagnosis and localization of faults in photovoltaic panel strings and bypass diodes. In: 2024 6th Int. Youth Conf. Radio Electron. Electr. Power Eng., IEEE; 2024, p. 1–7. https://doi.org/10.1109/REEPE60449.2024.10479707.
[232] Menemmeche H, Abderrezek M, Ahriche A. Design and development of cleaning solar plants robots. Energy Sources, Part A Recover Util Environ Eff 2023;45: 7031–49. https://doi.org/10.1080/15567036.2023.2218309.
[233] Kamarudin MN, Rozali SM, Jamri MS. Active cooling photovoltaic with IoT facility. Int J Power Electron Drive Syst 2021;12:1494. https://doi.org/10.11591/ijpeds.v12.i3.pp1494-1504.
[234] Taqwa A, Dewi T, Kusumanto R, Sitompul CR, Rusdianasari. Automatic cooling of a PV system to overcome overheated PV surface in Palembang. J Phys Conf Ser 2020;1500:012013. https://doi.org/10.1088/1742-6596/1500/1/012013.
[235] Laseinde OT, Ramere MD. Efficiency improvement in polycrystalline solar panel using thermal control water spraying cooling. Procedia Comput Sci 2021;180: 239–48. https://doi.org/10.1016/j.procs.2021.01.161.
[236] Prasad AR, Shankar R, Patil CK, Karthick A, Kumar A, Rahim R. Performance enhancement of solar photovoltaic system for roof top garden. Environ Sci Pollut Res 2021;28:50017–27. https://doi.org/10.1007/s11356-021-14191-z.
[237] Benato A, Stoppato A, De Vanna F, Schiro F. Spraying cooling system for PV modules: experimental measurements for temperature trends assessment and system design feasibility. Designs 2021;5:25. https://doi.org/10.3390/designs5020025.
[238] Nader N, Al-Kouz W, Al-Dahidi S. Assessment of existing photovoltaic system with cooling and cleaning system: case study at Al-Khobar City. Processes 2019;8:9. https://doi.org/10.3390/pr8010009.
[239] Paul S, Mondol N, Rahman S, Easmin F. Performance improvement of solar photovoltaic panel using smart water cooling technology. Int Conf Mech Ind Energy Eng 2020;2020:1–6.
[240] Ashok Kumar L, Indragandhi V, Teekaraman Y, Kuppusamy R, Radhakrishnan A. Design and implementation of automatic water spraying system for solar photovoltaic module. Math Probl Eng 2022;2022:1–11. https://doi.org/10.1155/2022/7129610.
[241] Cho I, Kim H. Study on PV panel cooling system using IoT with ESS for preventing reduced efficiency of solar panel. IOP Conf Ser Earth Environ Sci 2019;342: 012006. https://doi.org/10.1088/1755-1315/342/1/012006.
[242] Kusuma VA, Aprillia H, Suprapto SS, Ramadhani MN, Firdaus AA, Putra DFU. Analysis of the effect of a microcontroller-based solar panel cooling system on temperature and power output. Int J Appl Power Eng 2023;12:119. https://doi.org/10.11591/ijape.v12.i2.pp119-125.
[243] Farag MM, Hamid AK, Hussein M. Adaptive cooling framework for Photovoltaic systems: A seasonal investigation under the terrestrial conditions of Sharjah, UAE. 7th Int Conf Renew ENERGY Gener Appl 2024.
[244] Manasrah A, Masoud M, Jaradat Y, Bevilacqua P. Investigation of a real-time dynamic model for a PV cooling system. Energies 2022;15:1836. https://doi.org/10.3390/en15051836.
[245] Li B, Hong W, Li H, Lan J, Zi J. Optimized energy distribution management in the nanofluid-assisted photovoltaic/thermal system via exergy efficiency analysis. Energy 2022;242:123018. https://doi.org/10.1016/j.energy.2021.123018.

[246] Kaneesamkandi Z, Rehman AU. Selection of a photovoltaic panel cooling technique using multi-criteria decision analysis. Appl Sci 2023;13:1949. https://doi.org/10.3390/app13031949.
[247] Pruthviraj U, Kashyap Y, Baxevanaki E, Kosmopoulos P. Solar photovoltaic hotspot inspection using unmanned aerial vehicle thermal images at a solar field in South India. Remote Sens 2023;15:1914. https://doi.org/10.3390/rs15071914.
[248] Jakhar S, Paliwal MK, Kumar M. Machine learning predictive models for optimal design of photovoltaic/thermal collector with nanofluids based geothermal cooling. Environ Prog Sustain Energy 2023. https://doi.org/10.1002/ep.14131.